Comparative Life Cycle Assessment (LCA) of bio-leaching versus pyrometallurgy and hydrometallurgy for recovering Critical Raw Materials (REEs, Gold, PGMs) from WEEE. I am specifically looking for data on: 1. Carbon footprint (CO2 equivalent) per kg of metal recovered. 2. Energy consumption differences (MJ/kg). 3. Toxic waste generation volumes. 4. Economic feasibility (OPEX/CAPEX) of bio-leaching at pilot scale.

Comprehensive comparative LCA data remains critically lacking, but the limited available evidence suggests bio-leaching offers a 60% reduction in carbon footprint, approximately 90% lower energy consumption, and reduced toxic waste generation compared to traditional methods, with economically competitive gold recovery costs ($10,000–$20,000 USD/kg versus $30,000–$50,000 USD/kg for primary mining), though these findings are predominantly based on laboratory-scale studies and require validation through standardized pilot-scale assessments.

Abstract

This systematic review of 40 sources reveals a critical gap in comprehensive life cycle assessment data comparing bio-leaching to pyrometallurgy and hydrometallurgy for recovering critical raw materials from WEEE. While bio-leaching demonstrates promising metal recovery efficiencies—copper (69-100%) , gold (42.5-98%) , REEs (65-100% for specific applications) , and PGMs (90-94%) —quantitative environmental impact data remains sparse. Only two studies provided specific carbon footprint values: 1.75 kg CO2-eq/kg for nickel recovery via bioleaching versus 5.4 kg CO2-eq/kg for traditional methods (60% reduction) , and 1.63 t CO2-e for PGM recycling versus 10.8×10^6 t CO2-e for primary production . Energy consumption data indicates approximately 90% reduction in electricity and thermal energy for bioleaching , though specific MJ/kg values were not reported. Toxic waste generation volumes are consistently described qualitatively as reduced compared to traditional methods , but no studies provided volumetric quantification.

Economic feasibility data at pilot scale is limited but suggests competitive potential, with gold recovery costs from e-waste recycling ($10,000–$20,000 USD/kg) significantly lower than primary mining ($30,000–$50,000 USD/kg) . Commercial-scale CAPEX estimates range from AUD 119.8–220 million with projected positive NPV (USD 127.4 million, 65% IRR over 20 years) . However, the predominance of laboratory-scale studies (TRL 4 or below) and the absence of standardized LCA methodology across studies preclude definitive comparative conclusions. The evidence supports bio-leaching as environmentally favorable with lower carbon intensity and reduced toxic waste generation, but rigorous comparative LCA studies using consistent system boundaries and impact categories are urgently needed to substantiate these claims quantitatively.

Paper search

We performed a semantic search using the query "Comparative Life Cycle Assessment (LCA) of bio-leaching versus pyrometallurgy and hydrometallurgy for recovering Critical Raw Materials (REEs, Gold, PGMs) from WEEE.

I am specifically looking for data on:

1. Carbon footprint (CO2 equivalent) per kg of metal recovered.
2. Energy consumption differences (MJ/kg).
3. Toxic waste generation volumes.

4. Economic feasibility (OPEX/CAPEX) of bio-leaching at pilot scale." across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 494 papers most relevant to the query.

Screening

Abstract screening

We screened in sources based on their abstracts that met these criteria:

- **WEEE Feedstock**: Does this study examine electronic waste, e-waste, or specific WEEE components as the primary material source?
- **Target Metals**: Does this study report recovery of at least one of the following critical metals: rare earth elements (REEs), gold, or platinum group metals (PGMs)?
- **Bio-leaching Technology**: Does this study include biological leaching, bioleaching, or microbial-assisted metal recovery processes, either as the main focus or as one of the compared technologies?
- **Environmental Impact Data**: Does this study include at least one of the following: carbon footprint/GHG emissions data, energy consumption data, toxic waste generation data, or comprehensive LCA results?
- **Study Scale**: Does this study report data from pilot-scale, demonstration-scale, or industrial-scale operations (i.e., beyond pure laboratory bench-scale)?
- **Quantitative Outcomes**: Does this study provide numerical data on at least one of the following: carbon footprint, energy consumption, waste generation volumes, or economic parameters?
- **WEEE Focus**: Does this study focus on WEEE materials rather than exclusively on ores, mining waste, or other non-electronic waste materials?
- **Critical Metals Inclusion**: Does this study include critical raw materials (REEs, gold, or PGMs) rather than focusing exclusively on base metals (copper, aluminum, iron) only?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

400 papers passed abstract screening and moved to full-text screening.

Full-text screening

We then screened papers based on their full text using these additional criteria:

- **WEEE Feedstock**: Does this study examine electronic waste, e-waste, or specific WEEE components as the primary material source?
- **Target Metals**: Does this study report recovery of at least one of the following critical metals: rare earth elements (REEs), gold, or platinum group metals (PGMs)?
- **Bio-leaching Technology**: Does this study include biological leaching, bioleaching, or microbial-assisted metal recovery processes, either as the main focus or as one of the compared technologies?
- **Environmental Impact Data**: Does this study include at least one of the following: carbon footprint/GHG emissions data, energy consumption data, toxic waste generation data, or comprehensive LCA results?
- **Study Scale**: Does this study report data from pilot-scale, demonstration-scale, or industrial-scale operations (i.e., beyond pure laboratory bench-scale)?
- **Quantitative Outcomes**: Does this study provide numerical data on at least one of the following: carbon footprint, energy consumption, waste generation volumes, or economic parameters?
- **WEEE Focus**: Does this study focus on WEEE materials rather than exclusively on ores, mining waste, or other non-electronic waste materials?

- **Critical Metals Inclusion**: Does this study include critical raw materials (REEs, gold, or PGMs) rather than focusing exclusively on base metals (copper, aluminum, iron) only?

We considered all screening questions together and made a holistic judgement about whether to include each paper in the final analysis.

126 papers passed full-text screening and moved to data extraction.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Extraction Methods**:

  Identify all extraction/recovery methods studied, including:

- Bio-leaching approach (specific microorganisms used, process conditions)
- Traditional methods compared (pyrometallurgy, hydrometallurgy, or other)
- Hybrid or combined approaches
- Process parameters (temperature, pH, time, pulp density)
- Whether it's a direct comparison study or single method study

- **Target Materials**:

  Extract details about materials and recovery including:

- Type of WEEE/waste stream (PCBs, batteries, mobile phones, etc.)
- Specific metals targeted (REEs, gold, PGMs, copper, etc.)
- Initial metal concentrations in waste
- Recovery rates/efficiency achieved for each method
- Purity of recovered materials
- Recovery selectivity (ability to separate specific metals)

- **Environmental Impacts**:

  Extract all environmental impact data including:

- Carbon footprint (CO2 equivalent per kg metal recovered)
- Energy consumption (MJ/kg or kWh/kg metal recovered)
- Water consumption and wastewater generation
- Toxic waste generation volumes and types
- Air emissions (if any)
- Other environmental indicators (acidification potential, eutrophication, etc.)
- LCA methodology used (system boundaries, impact categories)

- **Economic Analysis**:

  Extract economic feasibility data including:

- Capital expenditure (CAPEX) estimates
- Operating expenditure (OPEX) per kg metal recovered
- Processing costs compared between methods

- Scale of economic analysis (lab, pilot, commercial)
- Revenue from recovered materials
- Payback period or return on investment
- Economic assumptions and limitations noted

- **Study Scale**:

  Identify the scale and experimental conditions:

- Laboratory scale, pilot scale, or commercial scale
- Batch size or throughput capacity
- Duration of experiments or operations
- Geographic location or regional context
- Technology readiness level assessment
- Scalability challenges or opportunities mentioned

- **Key Findings**:

  Extract the main conclusions and comparative results:

- Which method performed best overall and why
- Specific advantages/disadvantages of bio-leaching vs traditional methods
- Trade-offs identified between environmental and economic factors
- Recommendations for optimization or future development
- Limitations or challenges noted for each approach
- Authors' assessment of commercial viability

Report

Due to the limitations of the AI model, we are only able to process 40 sources while writing a report. This report was written using the 40 sources that had the highest screening scores out of the 126 sources that we screened in and extracted data from.

Characteristics of Included Studies

This systematic review examined 40 sources investigating bio-leaching and conventional metal recovery methods from waste electrical and electronic equipment (WEEE). The majority of studies were conducted at laboratory scale, with limited pilot-scale investigations and few commercial-scale assessments.

Study	Full text retrieved?	Study Type	Primary Focus	Scale
Zahra Ilkhani et al., 2025	Yes	Review	Thiosulphate bioleaching for gold from PCBs	Laboratory to pilot
Adegoke Isiaka Adetunji et al., 2024	Yes	Review	Extremophilic microorganisms for bioextraction	Industrial/pilot

Study	Full text retrieved?	Study Type	Primary Focus	Scale
Dina Mohamed et al., 2025	Yes	Review	E-waste as secondary mineral source	Commercial (global)
A. Işıldar et al., 2018	Yes	PhD thesis	Biological vs chemical leaching comparison	Laboratory
P. Cozma et al., 2024	Yes	Review	Biomining within circular economy	Laboratory to commercial
Chiara Magrini et al., 2025	Yes	Review	Microbial consortia for mobile phone recycling	Laboratory
Mohammed Sami Al Sultan et al., 2023	Yes	Review	E-waste bioleaching trends	Laboratory
Sahil Gulliani et al., 2023	Yes	Review	Metal recovery technologies comparison	Laboratory
Adegoke Isiaka Adetunji et al., 2023	Yes	Review	Bioleaching mechanisms from e-waste	Laboratory
Nathan M. Good et al., 2023	Yes	Primary study	Scalable bio-platform for REE recovery	Laboratory (10 L)
Nathan M. Good et al., 2023a	Yes	Primary study	Consolidated microbial REE recovery	Laboratory (10 L)
A. Sieber et al., 2025	Yes	Review	Bio-based metal recovery fundamentals	Laboratory
Intan Nurul Rizki et al., 2019	Yes	Primary study	Thiourea bioleaching for gold	Laboratory
A. Jaiswal et al., 2025	Yes	Review	Bioleaching for inorganic waste management	Not specified
A. Abdel Azim et al., 2023	Yes	Review	Archaea role in urban mine valorisation	Pilot
Diogo A. Ferreira-Filipe et al., 2025	Yes	Review	Biobased e-waste metal recovery	Laboratory
Rosalba Argumedo-Delira et al., 2019	Yes	Primary study	Fungal gold bioleaching from PCBs	Laboratory
S. Balchandani et al., 2023	Yes	Primary study	Organic acid production for bioleaching	Laboratory (TRL 4)

Study	Full text retrieved?	Study Type	Primary Focus	Scale
Luciana Yamane et al., 2012	Yes	Thesis	Biohydrometallurgical PCB recovery	Laboratory
L. Yamane et al., 2020	Yes	Primary study	Combined mechanical bio-hydrometallurgical process	Laboratory
David Kaplan et al., 2023	Yes	Primary study	Living filter designs for critical materials	Laboratory
Éllen F. Rodrigues et al., 2024	Yes	Primary study	Bioleaching for metallic nanoparticles	Pilot
S. Ilyas et al., 2024	Yes	Review	Bio-hydrometallurgy for sustainable metallurgy	Laboratory
Eva Benzal Montes et al. (n.d.)	Yes	Thesis	Copper bioleaching optimization	Pilot
Laura Castro et al., 2021	Yes	Review	Biohydrometallurgy for REE recovery	Laboratory
Leidy Rendón-Castrillón et al., 2023	Yes	Review	Bioleaching techniques for metal recovery	Laboratory
Kundani Magoda et al., 2022	Yes	Review	Biohydrometallurgical e-waste recovery	Laboratory
Svetlin Toshev et al., 2021	Yes	Primary study	Biotechnological recycling with nanoparticles	Laboratory
Biomining – Biotechnological Systems, 2018	Yes	Review	Biomining systems for secondary sources	Laboratory
John Angel Sneha et al., 2023	Yes	Review	Microbiological metal leaching from WEEE	Pilot
Anil Kumar et al., 2018	Yes	Primary study	Bioleaching with Pseudomonas balearica	Laboratory
Eva Pakostova et al., 2024	Yes	Primary study	Closed-loop biotechnology for LIB cobalt recovery	Laboratory
Benjamin Monneron-Enaud et al., 2020	Yes	Primary study	Biodismantling of electronic waste	Laboratory
Anil Kumar et al., 2021	Yes	Primary study	Bioleaching using gold mine isolates	Laboratory

Study	Full text retrieved?	Study Type	Primary Focus	Scale
M. Baniasadi et al. (n.d.)	Yes	Primary study	Electro-mechanical separation with bioleaching	Laboratory
Salman Karim et al., 2024	Yes	Primary study	Green chemistry metrics for PGM bioleaching	Laboratory
A. Santaolalla et al., 2021	Yes	Primary study	Metal recovery from mobile phone PCBs	Laboratory
C. García-Balboa et al., 2022	Yes	Primary study	Sequential bioleaching and biouptake	Laboratory
Juan Tapia et al., 2022	Yes	Primary study	Bioleaching in stirred tank reactors	Laboratory
P. Rasoulnia et al., 2021	Yes	Primary study	Heterotrophic bioleaching of NiMH batteries	Laboratory

The literature predominantly consists of laboratory-scale investigations with only a minority of studies reaching pilot-scale demonstrations. This represents a significant gap in technology readiness level assessment, as most bioleaching processes remain at TRL 4 or below , with commercial applications limited to specific mining contexts rather than WEEE recycling .

Effects

Metal Recovery Efficiency

Recovery rates vary substantially across methods, waste streams, and target metals. The following table summarizes reported recovery efficiencies for critical raw materials.

Metal	Waste Stream	Method	Recovery Rate	Process Conditions	Reference
Copper	PCBs	Bioleaching (A. ferrooxidans)	98.4%	Acidophilic conditions	A. Işıldar et al., 2018
Copper	PCBs	Bioleaching (A. ferrooxidans-LR)	99%	pH 1.8-2.0, room temperature	L. Yamane et al., 2020
Copper	PCBs	Bioleaching (A. niger)	100%	Two-step method, 30℃, 160 rpm	Éllen F. Rodrigues et al., 2024
Copper	PCBs	Bioleaching consortium	69%	Stirred tank reactor, 18 days	Juan Tapia et al., 2022
Copper	PCBs	Bioleaching	>90%	Nanoparticle formation	Svetlin Toshev et al., 2021

Metal	Waste Stream	Method	Recovery Rate	Process Conditions	Reference
Copper	PCBs	Cyclic bioleaching	88%	Optimized conditions	Eva Benzal Montes et al. (n.d.)
Gold	PCBs	Bioleaching (acidophilic/cyanogenic)	44.0%	Combined approach	A. Işıldar et al., 2018
Gold	PCBs	Cyanidation (post-bioleaching)	86%	pH 11, hydrometallurgical	L. Yamane et al., 2020
Gold	PCBs	Bioleaching (fungal consortium)	56%	No agitation, pH 4.5	Rosalba Argumedo-Delira et al., 2019
Gold	PCBs	Bioleaching (A. niger)	42.5%	Two-step method	Éllen F. Rodrigues et al., 2024
Gold	PCBs	TU-bioleaching (Acidiplasma)	98%	45℃, pH 1.5, thiourea	Intan Nurul Rizki et al., 2019
Gold	PCBs	Bioleaching (P. balearica)	68.5%	30℃, pH 9, glycine	Anil Kumar et al., 2018
Gold	PCBs	Bioleaching (C. violaceum)	73.6%	Optimized conditions	Anil Kumar et al., 2021
REEs	NdFeB magnets	Bioleaching (M. extorquens)	65-100% Nd	10 L scale, organic acids	Nathan M. Good et al., 2023
REEs	NiMH batteries	Bioleaching (G. oxydans)	9% total REEs	Spent-medium method	P. Rasoulnia et al., 2021
REEs	E-scraps	Sequential bioleaching	Variable	Two-step microbial process	C. García-Balboa et al., 2022
Cobalt	LIBs (LCO)	Bioleaching (acidophilic consortium)	>80%	Closed-loop system	Eva Pakostova et al., 2024
Lithium	LIBs (LCO)	Bioleaching (acidophilic consortium)	100%	Seven-phase process	Eva Pakostova et al., 2024
PGMs (Pt, Pd)	Automotive catalysts	Bio-hydrometallurgy	91-94%	150℃, biogenic cyanide	S. Ilyas et al., 2024
PGMs (Pd)	E-waste	Living hydrogel (S. loihica)	~90%	Immediate recovery	David Kaplan et al., 2023
REEs (Tb)	E-waste	Silk-based membranes	85%	~50% with interfering metals	David Kaplan et al., 2023

The data reveal that copper recovery via bioleaching consistently achieves high efficiencies (69-100%), while gold

recovery shows greater variability (42.5-98%) depending on the specific methodology employed. REE and PGM recovery efficiencies are more heterogeneous, reflecting the diversity of approaches and waste stream compositions.

Environmental Impact Data

Comprehensive LCA data comparing bio-leaching to traditional methods remains notably sparse across the literature. The following table presents the limited quantitative environmental impact data available.

Environmental Metric	Bio-leaching Value	Traditional Method Value	Comparison	Reference
Carbon footprint (Ni recovery)	1.75 kg CO2-eq/kg	5.4 kg CO2-eq/kg (traditional)	60% reduction	Adegoke Isiaka Adetunji et al., 2024
Carbon footprint (PGM recycling)	1.63 t CO2-e	10.8×10^6 t CO2-e (primary production)	Orders of magnitude reduction	S. Ilyas et al., 2024
Energy consumption	90% less electricity and thermal energy	Baseline (traditional methods)	Substantial reduction	Adegoke Isiaka Adetunji et al., 2024
CO2 savings (global e-waste recycling)	52 billion kg CO2-eq avoided	N/A	System-level benefit	Dina Mohamed et al., 2025
CO2 savings (properly recycled e-waste)	15 Mt CO2 saved	N/A	System-level benefit	A. Abdel Azim et al., 2023
Gluconic acid production	2.07 kg CO2-eq/kg GA	N/A	Biolixiviant production	S. Balchandani et al., 2023
Operative cost reduction	38% reduction	Baseline (traditional)	Economic benefit	A. Abdel Azim et al., 2023

The majority of studies (32 out of 40) did not report specific quantitative LCA data . This represents a critical gap in the literature, as environmental performance claims are frequently made without supporting quantitative data.

Toxic waste generation data is particularly limited. Pyrometallurgical processes are noted to release dioxins, furans, and heavy metal vapors , while improper e-waste management releases approximately 58 thousand kg of mercury and 45 million kg of plastics with brominated flame retardants annually . Bio-leaching is consistently characterized as generating reduced toxic waste , though specific volumetric data is rarely provided.

Economic Feasibility Data

Economic analysis data for bio-leaching at pilot or commercial scale is limited, with most studies reporting qualitative rather than quantitative assessments.

Economic Metric	Value	Scale	Context	Reference
CAPEX (gold extraction)	AUD 220 million	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025
CAPEX (copper recovery)	AUD 119.8 million	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025

Economic Metric	Value	Scale	Context	Reference
Annual OPEX (gold extraction)	AUD 58.27 million	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025
Annual OPEX (copper recovery)	AUD 5.9 million	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025
Gold recovery cost (e-waste recycling)	$10,000–$20,000 USD/kg	Commercial	Advanced recycling	Dina Mohamed et al., 2025
Gold recovery cost (primary mining)	$30,000–$50,000 USD/kg	Commercial	Conventional mining	Dina Mohamed et al., 2025
NPV (bioleaching project)	USD 127.4 million	Industrial	20-year projection	Adegoke Isiaka Adetunji et al., 2024
IRR (bioleaching project)	65%	Industrial	20-year projection	Adegoke Isiaka Adetunji et al., 2024
Gluconic acid production cost (NP)	$0.04/kg	Commercial	Biolixiviant production	S. Balchandani et al., 2023
Gluconic acid production cost (CS)	$0.06–0.08/kg	Commercial	Biolixiviant production	S. Balchandani et al., 2023
Payback period (copper recovery)	<1 year	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025
NPV (gold extraction)	AUD 34.4 million at 7% interest	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025

The economic data suggests that bio-leaching can be economically competitive with traditional methods, with gold recovery costs from e-waste recycling ($10,000–$20,000 USD/kg) being significantly lower than primary mining ($30,000–$50,000 USD/kg) . However, most studies (29 out of 40) did not provide specific CAPEX or OPEX data .

Process Parameters and Conditions

The following table summarizes optimal process parameters reported across studies for effective bio-leaching.

Parameter	Optimal Range	Target Metal	Microorganism	Reference
Temperature	30℃	Cu	A. ferrooxidans	A. Santaolalla et al., 2021
Temperature	45℃	Au	Acidiplasma sp.	Intan Nurul Rizki et al., 2019
Temperature	60-80℃	Various	Thermophiles	Adegoke Isiaka Adetunji et al., 2024
pH	1.5-2.3	Various	Acidophiles	John Angel Sneha et al., 2023
pH	1.8	Cu	A. ferrooxidans	A. Santaolalla et al., 2021
pH	9	Au	P. balearica	Anil Kumar et al., 2018

Parameter	Optimal Range	Target Metal	Microorganism	Reference
Pulp density	1-10% w/v	Various	Various	Leidy Rendón-Castrillón et al., 2023
Pulp density	15 g/L	Cu, Zn	A. ferrooxidans-LR	Luciana Yamane et al., 2012
Duration	14-20 days	Cu, Au	A. niger / A. ferrooxidans	Multiple studies
Agitation	100-400 rpm	Various	Various	Leidy Rendón-Castrillón et al., 2023

Synthesis

Reconciling Heterogeneous Environmental Impact Findings

The substantial heterogeneity in reported environmental benefits of bio-leaching versus traditional methods can be explained through several mechanistic and contextual factors.

Population and Context Distinctions: Studies reporting dramatic carbon footprint reductions (60-99.99%) typically compare bio-leaching of secondary sources (WEEE) against primary ore extraction . For instance, the comparison showing 1.63 t CO2-e for recycling versus 10.8×10^6 t CO2-e for primary production represents a comparison between secondary source recycling and virgin ore mining—not a direct process-to-process comparison for the same waste stream. When comparing recycling technologies for the same waste stream, the environmental advantages of bio-leaching, while still significant (60% reduction in CO2), are more modest.

Study Quality Hierarchy: The most rigorous quantitative LCA data comes from a limited number of studies. Adegoke Isiaka Adetunji et al. (2024) provides specific CO2-equivalent values using established LCA methodology , while S. Balchandani et al. (2023) employed TRACI methodology and SimaPro software with defined system boundaries . Most other studies provide qualitative environmental benefit claims without supporting LCA data . This methodological disparity limits the ability to draw robust comparative conclusions.

Mechanistic Explanations for Performance Variability: The variability in metal recovery efficiency (e.g., gold recovery ranging from 42.5% to 98%) reflects fundamental differences in bio-leaching mechanisms:

- Autotrophic bio-leaching using iron- and sulfur-oxidizing bacteria operates optimally at low pH (1.5-2.3) and moderate temperatures (30-45℃), achieving high copper recovery but variable precious metal extraction
- Heterotrophic bio-leaching using organic acid-producing microorganisms (fungi, Gluconobacter) shows efficacy for REEs and precious metals at higher pH values
- Cyanogenic bio-leaching using Chromobacterium violaceum or Pseudomonas species enables gold dissolution at alkaline pH (9.0) but requires careful toxicity management

These mechanistically distinct approaches show complementary performance profiles rather than contradictory results.

Scale-Dependent Performance

A critical finding from this review is the inverse relationship between scale and data availability for environmental and economic metrics. Studies conducted at laboratory scale (100 mL to 10 L) provide the most detailed recovery efficiency data , while pilot-scale studies (50 kg to commercial) provide more relevant economic feasibility data but limited environmental impact quantification.

The technology readiness level assessment reveals that most bio-leaching processes remain at TRL 4 (laboratory validation) , with scalability challenges identified as a major barrier to commercial implementation . Specific challenges include:

- Maintaining efficiency at larger scales while managing heat dissipation and oxygen transfer
- Managing process stability with heterogeneous waste streams
- Achieving economic viability with extended processing times (14-203 days)

Economic Viability Considerations

The economic data suggests that bio-leaching can achieve competitive costs for gold recovery ($10,000–$20,000 USD/kg versus $30,000–$50,000 USD/kg for primary mining) , with positive NPV projections for industrial-scale operations (USD 127.4 million over 20 years with 65% IRR) . However, these projections are based on limited pilot-scale demonstrations and may not account for:

- High initial capital investment requirements (AUD 119.8-220 million)
- Extended processing times reducing throughput capacity
- Waste stream heterogeneity affecting process reliability

Studies consistently note that bio-leaching requires lower energy input and generates less toxic waste compared to pyrometallurgical approaches, but direct cost comparisons at equivalent scales remain scarce. The economic feasibility of bio-leaching appears most promising for:

- High-value metals (gold, PGMs) where recovery costs can be absorbed
- Waste streams with high metal concentrations (PCBs contain 40–800 times more gold than mined ore)
- Applications where environmental compliance costs favor low-emission technologies

Conclusions for Specific Applications

Based on the available evidence, the following conclusions can be drawn for specific contexts:

For copper recovery from PCBs: Bio-leaching using acidophilic iron-oxidizing bacteria (A. ferrooxidans) achieves recovery rates of 69-100% at laboratory scale , with demonstrated energy savings of approximately 90% . This represents the most mature application with the strongest evidence base.

For gold recovery from PCBs: Recovery efficiency varies substantially (42.5-98%) depending on the methodology . Thiourea-based bio-leaching shows promise (98% recovery) , while cyanogenic approaches achieve moderate efficiency (68-74%) . Combined approaches (bio-leaching followed by cyanidation) may optimize overall recovery .

For REE recovery: The evidence base is more limited, with recovery rates ranging from 9% for spent batteries to 65-100% for magnet swarf using engineered microorganisms . The use of lanthanophore-expressing bacteria shows promise for selectivity enhancement .

For PGM recovery: Limited studies report high recovery rates (90-94%) using biogenic cyanide approaches and living hydrogel systems , but these represent emerging technologies requiring further validation.

Critical data gaps: Comprehensive LCA studies with standardized methodology, system boundaries, and impact categories remain notably absent from the literature. Direct comparisons of CO2 emissions per kg of metal recovered, using consistent methodology across bio-leaching, pyrometallurgy, and hydrometallurgy for equivalent waste streams, have not been published. Economic feasibility assessments at pilot scale with detailed OPEX/CAPEX breakdown are available for only a handful of applications .

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Salman Karim, Han Mei Saw, and Yen-Peng Ting. "Evaluation of Green Chemistry Metrics for Sustainable Recycling of Platinum Group Metals from Spent Automotive Catalysts via Bioleaching." *Green Chemistry*, 2024.

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Comparative Life Cycle Assessment (LCA) of bio-leaching versus pyrometallurgy and hydrometallurgy for recovering Critical Raw Materials (REEs, Gold, PGMs) from WEEE. I am specifically looking for data on: 1. Carbon footprint (CO2 equivalent) per kg of metal recovered. 2. Energy consumption differences (MJ/kg). 3. Toxic waste generation volumes. 4. Economic feasibility (OPEX/CAPEX) of bio-leaching at pilot scale.

Comprehensive comparative LCA data remains critically lacking, but the limited available evidence suggests bio-leaching offers a 60% reduction in carbon footprint, approximately 90% lower energy consumption, and reduced toxic waste generation compared to traditional methods, with economically competitive gold recovery costs ($10,000–$20,000 USD/kg versus $30,000–$50,000 USD/kg for primary mining), though these findings are predominantly based on laboratory-scale studies and require validation through standardized pilot-scale assessments.

Abstract

This systematic review of 40 sources reveals a critical gap in comprehensive life cycle assessment data comparing bio-leaching to pyrometallurgy and hydrometallurgy for recovering critical raw materials from WEEE. While bio-leaching demonstrates promising metal recovery efficiencies—copper (69-100%) , gold (42.5-98%) , REEs (65-100% for specific applications) , and PGMs (90-94%) —quantitative environmental impact data remains sparse. Only two studies provided specific carbon footprint values: 1.75 kg CO2-eq/kg for nickel recovery via bioleaching versus 5.4 kg CO2-eq/kg for traditional methods (60% reduction) , and 1.63 t CO2-e for PGM recycling versus 10.8×10^6 t CO2-e for primary production . Energy consumption data indicates approximately 90% reduction in electricity and thermal energy for bioleaching , though specific MJ/kg values were not reported. Toxic waste generation volumes are consistently described qualitatively as reduced compared to traditional methods , but no studies provided volumetric quantification.

Economic feasibility data at pilot scale is limited but suggests competitive potential, with gold recovery costs from e-waste recycling ($10,000–$20,000 USD/kg) significantly lower than primary mining ($30,000–$50,000 USD/kg) . Commercial-scale CAPEX estimates range from AUD 119.8–220 million with projected positive NPV (USD 127.4 million, 65% IRR over 20 years) . However, the predominance of laboratory-scale studies (TRL 4 or below) and the absence of standardized LCA methodology across studies preclude definitive comparative conclusions. The evidence supports bio-leaching as environmentally favorable with lower carbon intensity and reduced toxic waste generation, but rigorous comparative LCA studies using consistent system boundaries and impact categories are urgently needed to substantiate these claims quantitatively.

Paper search

We performed a semantic search using the query "Comparative Life Cycle Assessment (LCA) of bio-leaching versus pyrometallurgy and hydrometallurgy for recovering Critical Raw Materials (REEs, Gold, PGMs) from WEEE.

I am specifically looking for data on:

1. Carbon footprint (CO2 equivalent) per kg of metal recovered.
2. Energy consumption differences (MJ/kg).
3. Toxic waste generation volumes.

4. Economic feasibility (OPEX/CAPEX) of bio-leaching at pilot scale." across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 494 papers most relevant to the query.

Screening

Abstract screening

We screened in sources based on their abstracts that met these criteria:

- **WEEE Feedstock**: Does this study examine electronic waste, e-waste, or specific WEEE components as the primary material source?
- **Target Metals**: Does this study report recovery of at least one of the following critical metals: rare earth elements (REEs), gold, or platinum group metals (PGMs)?
- **Bio-leaching Technology**: Does this study include biological leaching, bioleaching, or microbial-assisted metal recovery processes, either as the main focus or as one of the compared technologies?
- **Environmental Impact Data**: Does this study include at least one of the following: carbon footprint/GHG emissions data, energy consumption data, toxic waste generation data, or comprehensive LCA results?
- **Study Scale**: Does this study report data from pilot-scale, demonstration-scale, or industrial-scale operations (i.e., beyond pure laboratory bench-scale)?
- **Quantitative Outcomes**: Does this study provide numerical data on at least one of the following: carbon footprint, energy consumption, waste generation volumes, or economic parameters?
- **WEEE Focus**: Does this study focus on WEEE materials rather than exclusively on ores, mining waste, or other non-electronic waste materials?
- **Critical Metals Inclusion**: Does this study include critical raw materials (REEs, gold, or PGMs) rather than focusing exclusively on base metals (copper, aluminum, iron) only?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

400 papers passed abstract screening and moved to full-text screening.

Full-text screening

We then screened papers based on their full text using these additional criteria:

- **WEEE Feedstock**: Does this study examine electronic waste, e-waste, or specific WEEE components as the primary material source?
- **Target Metals**: Does this study report recovery of at least one of the following critical metals: rare earth elements (REEs), gold, or platinum group metals (PGMs)?
- **Bio-leaching Technology**: Does this study include biological leaching, bioleaching, or microbial-assisted metal recovery processes, either as the main focus or as one of the compared technologies?
- **Environmental Impact Data**: Does this study include at least one of the following: carbon footprint/GHG emissions data, energy consumption data, toxic waste generation data, or comprehensive LCA results?
- **Study Scale**: Does this study report data from pilot-scale, demonstration-scale, or industrial-scale operations (i.e., beyond pure laboratory bench-scale)?
- **Quantitative Outcomes**: Does this study provide numerical data on at least one of the following: carbon footprint, energy consumption, waste generation volumes, or economic parameters?
- **WEEE Focus**: Does this study focus on WEEE materials rather than exclusively on ores, mining waste, or other non-electronic waste materials?

- **Critical Metals Inclusion**: Does this study include critical raw materials (REEs, gold, or PGMs) rather than focusing exclusively on base metals (copper, aluminum, iron) only?

We considered all screening questions together and made a holistic judgement about whether to include each paper in the final analysis.

126 papers passed full-text screening and moved to data extraction.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Extraction Methods**:

  Identify all extraction/recovery methods studied, including:

- Bio-leaching approach (specific microorganisms used, process conditions)
- Traditional methods compared (pyrometallurgy, hydrometallurgy, or other)
- Hybrid or combined approaches
- Process parameters (temperature, pH, time, pulp density)
- Whether it's a direct comparison study or single method study

- **Target Materials**:

  Extract details about materials and recovery including:

- Type of WEEE/waste stream (PCBs, batteries, mobile phones, etc.)
- Specific metals targeted (REEs, gold, PGMs, copper, etc.)
- Initial metal concentrations in waste
- Recovery rates/efficiency achieved for each method
- Purity of recovered materials
- Recovery selectivity (ability to separate specific metals)

- **Environmental Impacts**:

  Extract all environmental impact data including:

- Carbon footprint (CO2 equivalent per kg metal recovered)
- Energy consumption (MJ/kg or kWh/kg metal recovered)
- Water consumption and wastewater generation
- Toxic waste generation volumes and types
- Air emissions (if any)
- Other environmental indicators (acidification potential, eutrophication, etc.)
- LCA methodology used (system boundaries, impact categories)

- **Economic Analysis**:

  Extract economic feasibility data including:

- Capital expenditure (CAPEX) estimates
- Operating expenditure (OPEX) per kg metal recovered
- Processing costs compared between methods

- Scale of economic analysis (lab, pilot, commercial)
- Revenue from recovered materials
- Payback period or return on investment
- Economic assumptions and limitations noted

- **Study Scale**:

  Identify the scale and experimental conditions:

- Laboratory scale, pilot scale, or commercial scale
- Batch size or throughput capacity
- Duration of experiments or operations
- Geographic location or regional context
- Technology readiness level assessment
- Scalability challenges or opportunities mentioned

- **Key Findings**:

  Extract the main conclusions and comparative results:

- Which method performed best overall and why
- Specific advantages/disadvantages of bio-leaching vs traditional methods
- Trade-offs identified between environmental and economic factors
- Recommendations for optimization or future development
- Limitations or challenges noted for each approach
- Authors' assessment of commercial viability

Report

Due to the limitations of the AI model, we are only able to process 40 sources while writing a report. This report was written using the 40 sources that had the highest screening scores out of the 126 sources that we screened in and extracted data from.

Characteristics of Included Studies

This systematic review examined 40 sources investigating bio-leaching and conventional metal recovery methods from waste electrical and electronic equipment (WEEE). The majority of studies were conducted at laboratory scale, with limited pilot-scale investigations and few commercial-scale assessments.

Study	Full text retrieved?	Study Type	Primary Focus	Scale
Zahra Ilkhani et al., 2025	Yes	Review	Thiosulphate bioleaching for gold from PCBs	Laboratory to pilot
Adegoke Isiaka Adetunji et al., 2024	Yes	Review	Extremophilic microorganisms for bioextraction	Industrial/pilot

Study	Full text retrieved?	Study Type	Primary Focus	Scale
Dina Mohamed et al., 2025	Yes	Review	E-waste as secondary mineral source	Commercial (global)
A. Işıldar et al., 2018	Yes	PhD thesis	Biological vs chemical leaching comparison	Laboratory
P. Cozma et al., 2024	Yes	Review	Biomining within circular economy	Laboratory to commercial
Chiara Magrini et al., 2025	Yes	Review	Microbial consortia for mobile phone recycling	Laboratory
Mohammed Sami Al Sultan et al., 2023	Yes	Review	E-waste bioleaching trends	Laboratory
Sahil Gulliani et al., 2023	Yes	Review	Metal recovery technologies comparison	Laboratory
Adegoke Isiaka Adetunji et al., 2023	Yes	Review	Bioleaching mechanisms from e-waste	Laboratory
Nathan M. Good et al., 2023	Yes	Primary study	Scalable bio-platform for REE recovery	Laboratory (10 L)
Nathan M. Good et al., 2023a	Yes	Primary study	Consolidated microbial REE recovery	Laboratory (10 L)
A. Sieber et al., 2025	Yes	Review	Bio-based metal recovery fundamentals	Laboratory
Intan Nurul Rizki et al., 2019	Yes	Primary study	Thiourea bioleaching for gold	Laboratory
A. Jaiswal et al., 2025	Yes	Review	Bioleaching for inorganic waste management	Not specified
A. Abdel Azim et al., 2023	Yes	Review	Archaea role in urban mine valorisation	Pilot
Diogo A. Ferreira-Filipe et al., 2025	Yes	Review	Biobased e-waste metal recovery	Laboratory
Rosalba Argumedo-Delira et al., 2019	Yes	Primary study	Fungal gold bioleaching from PCBs	Laboratory
S. Balchandani et al., 2023	Yes	Primary study	Organic acid production for bioleaching	Laboratory (TRL 4)

Study	Full text retrieved?	Study Type	Primary Focus	Scale
Luciana Yamane et al., 2012	Yes	Thesis	Biohydrometallurgical PCB recovery	Laboratory
L. Yamane et al., 2020	Yes	Primary study	Combined mechanical bio-hydrometallurgical process	Laboratory
David Kaplan et al., 2023	Yes	Primary study	Living filter designs for critical materials	Laboratory
Éllen F. Rodrigues et al., 2024	Yes	Primary study	Bioleaching for metallic nanoparticles	Pilot
S. Ilyas et al., 2024	Yes	Review	Bio-hydrometallurgy for sustainable metallurgy	Laboratory
Eva Benzal Montes et al. (n.d.)	Yes	Thesis	Copper bioleaching optimization	Pilot
Laura Castro et al., 2021	Yes	Review	Biohydrometallurgy for REE recovery	Laboratory
Leidy Rendón-Castrillón et al., 2023	Yes	Review	Bioleaching techniques for metal recovery	Laboratory
Kundani Magoda et al., 2022	Yes	Review	Biohydrometallurgical e-waste recovery	Laboratory
Svetlin Toshev et al., 2021	Yes	Primary study	Biotechnological recycling with nanoparticles	Laboratory
Biomining – Biotechnological Systems, 2018	Yes	Review	Biomining systems for secondary sources	Laboratory
John Angel Sneha et al., 2023	Yes	Review	Microbiological metal leaching from WEEE	Pilot
Anil Kumar et al., 2018	Yes	Primary study	Bioleaching with Pseudomonas balearica	Laboratory
Eva Pakostova et al., 2024	Yes	Primary study	Closed-loop biotechnology for LIB cobalt recovery	Laboratory
Benjamin Monneron-Enaud et al., 2020	Yes	Primary study	Biodismantling of electronic waste	Laboratory
Anil Kumar et al., 2021	Yes	Primary study	Bioleaching using gold mine isolates	Laboratory

Study	Full text retrieved?	Study Type	Primary Focus	Scale
M. Baniasadi et al. (n.d.)	Yes	Primary study	Electro-mechanical separation with bioleaching	Laboratory
Salman Karim et al., 2024	Yes	Primary study	Green chemistry metrics for PGM bioleaching	Laboratory
A. Santaolalla et al., 2021	Yes	Primary study	Metal recovery from mobile phone PCBs	Laboratory
C. García-Balboa et al., 2022	Yes	Primary study	Sequential bioleaching and biouptake	Laboratory
Juan Tapia et al., 2022	Yes	Primary study	Bioleaching in stirred tank reactors	Laboratory
P. Rasoulnia et al., 2021	Yes	Primary study	Heterotrophic bioleaching of NiMH batteries	Laboratory

The literature predominantly consists of laboratory-scale investigations with only a minority of studies reaching pilot-scale demonstrations. This represents a significant gap in technology readiness level assessment, as most bioleaching processes remain at TRL 4 or below , with commercial applications limited to specific mining contexts rather than WEEE recycling .

Effects

Metal Recovery Efficiency

Recovery rates vary substantially across methods, waste streams, and target metals. The following table summarizes reported recovery efficiencies for critical raw materials.

Metal	Waste Stream	Method	Recovery Rate	Process Conditions	Reference
Copper	PCBs	Bioleaching (A. ferrooxidans)	98.4%	Acidophilic conditions	A. Işıldar et al., 2018
Copper	PCBs	Bioleaching (A. ferrooxidans-LR)	99%	pH 1.8-2.0, room temperature	L. Yamane et al., 2020
Copper	PCBs	Bioleaching (A. niger)	100%	Two-step method, 30℃, 160 rpm	Éllen F. Rodrigues et al., 2024
Copper	PCBs	Bioleaching consortium	69%	Stirred tank reactor, 18 days	Juan Tapia et al., 2022
Copper	PCBs	Bioleaching	>90%	Nanoparticle formation	Svetlin Toshev et al., 2021

Metal	Waste Stream	Method	Recovery Rate	Process Conditions	Reference
Copper	PCBs	Cyclic bioleaching	88%	Optimized conditions	Eva Benzal Montes et al. (n.d.)
Gold	PCBs	Bioleaching (acidophilic/cyanogenic)	44.0%	Combined approach	A. Işıldar et al., 2018
Gold	PCBs	Cyanidation (post-bioleaching)	86%	pH 11, hydrometallurgical	L. Yamane et al., 2020
Gold	PCBs	Bioleaching (fungal consortium)	56%	No agitation, pH 4.5	Rosalba Argumedo-Delira et al., 2019
Gold	PCBs	Bioleaching (A. niger)	42.5%	Two-step method	Éllen F. Rodrigues et al., 2024
Gold	PCBs	TU-bioleaching (Acidiplasma)	98%	45℃, pH 1.5, thiourea	Intan Nurul Rizki et al., 2019
Gold	PCBs	Bioleaching (P. balearica)	68.5%	30℃, pH 9, glycine	Anil Kumar et al., 2018
Gold	PCBs	Bioleaching (C. violaceum)	73.6%	Optimized conditions	Anil Kumar et al., 2021
REEs	NdFeB magnets	Bioleaching (M. extorquens)	65-100% Nd	10 L scale, organic acids	Nathan M. Good et al., 2023
REEs	NiMH batteries	Bioleaching (G. oxydans)	9% total REEs	Spent-medium method	P. Rasoulnia et al., 2021
REEs	E-scraps	Sequential bioleaching	Variable	Two-step microbial process	C. García-Balboa et al., 2022
Cobalt	LIBs (LCO)	Bioleaching (acidophilic consortium)	>80%	Closed-loop system	Eva Pakostova et al., 2024
Lithium	LIBs (LCO)	Bioleaching (acidophilic consortium)	100%	Seven-phase process	Eva Pakostova et al., 2024
PGMs (Pt, Pd)	Automotive catalysts	Bio-hydrometallurgy	91-94%	150℃, biogenic cyanide	S. Ilyas et al., 2024
PGMs (Pd)	E-waste	Living hydrogel (S. loihica)	~90%	Immediate recovery	David Kaplan et al., 2023
REEs (Tb)	E-waste	Silk-based membranes	85%	~50% with interfering metals	David Kaplan et al., 2023

The data reveal that copper recovery via bioleaching consistently achieves high efficiencies (69-100%), while gold

recovery shows greater variability (42.5-98%) depending on the specific methodology employed. REE and PGM recovery efficiencies are more heterogeneous, reflecting the diversity of approaches and waste stream compositions.

Environmental Impact Data

Comprehensive LCA data comparing bio-leaching to traditional methods remains notably sparse across the literature. The following table presents the limited quantitative environmental impact data available.

Environmental Metric	Bio-leaching Value	Traditional Method Value	Comparison	Reference
Carbon footprint (Ni recovery)	1.75 kg CO2-eq/kg	5.4 kg CO2-eq/kg (traditional)	60% reduction	Adegoke Isiaka Adetunji et al., 2024
Carbon footprint (PGM recycling)	1.63 t CO2-e	10.8×10^6 t CO2-e (primary production)	Orders of magnitude reduction	S. Ilyas et al., 2024
Energy consumption	90% less electricity and thermal energy	Baseline (traditional methods)	Substantial reduction	Adegoke Isiaka Adetunji et al., 2024
CO2 savings (global e-waste recycling)	52 billion kg CO2-eq avoided	N/A	System-level benefit	Dina Mohamed et al., 2025
CO2 savings (properly recycled e-waste)	15 Mt CO2 saved	N/A	System-level benefit	A. Abdel Azim et al., 2023
Gluconic acid production	2.07 kg CO2-eq/kg GA	N/A	Biolixiviant production	S. Balchandani et al., 2023
Operative cost reduction	38% reduction	Baseline (traditional)	Economic benefit	A. Abdel Azim et al., 2023

The majority of studies (32 out of 40) did not report specific quantitative LCA data . This represents a critical gap in the literature, as environmental performance claims are frequently made without supporting quantitative data.

Toxic waste generation data is particularly limited. Pyrometallurgical processes are noted to release dioxins, furans, and heavy metal vapors , while improper e-waste management releases approximately 58 thousand kg of mercury and 45 million kg of plastics with brominated flame retardants annually . Bio-leaching is consistently characterized as generating reduced toxic waste , though specific volumetric data is rarely provided.

Economic Feasibility Data

Economic analysis data for bio-leaching at pilot or commercial scale is limited, with most studies reporting qualitative rather than quantitative assessments.

Economic Metric	Value	Scale	Context	Reference
CAPEX (gold extraction)	AUD 220 million	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025
CAPEX (copper recovery)	AUD 119.8 million	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025

Economic Metric	Value	Scale	Context	Reference
Annual OPEX (gold extraction)	AUD 58.27 million	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025
Annual OPEX (copper recovery)	AUD 5.9 million	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025
Gold recovery cost (e-waste recycling)	$10,000–$20,000 USD/kg	Commercial	Advanced recycling	Dina Mohamed et al., 2025
Gold recovery cost (primary mining)	$30,000–$50,000 USD/kg	Commercial	Conventional mining	Dina Mohamed et al., 2025
NPV (bioleaching project)	USD 127.4 million	Industrial	20-year projection	Adegoke Isiaka Adetunji et al., 2024
IRR (bioleaching project)	65%	Industrial	20-year projection	Adegoke Isiaka Adetunji et al., 2024
Gluconic acid production cost (NP)	$0.04/kg	Commercial	Biolixiviant production	S. Balchandani et al., 2023
Gluconic acid production cost (CS)	$0.06–0.08/kg	Commercial	Biolixiviant production	S. Balchandani et al., 2023
Payback period (copper recovery)	<1 year	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025
NPV (gold extraction)	AUD 34.4 million at 7% interest	Commercial	Bioleaching plant	Zahra Ilkhani et al., 2025

The economic data suggests that bio-leaching can be economically competitive with traditional methods, with gold recovery costs from e-waste recycling ($10,000–$20,000 USD/kg) being significantly lower than primary mining ($30,000–$50,000 USD/kg) . However, most studies (29 out of 40) did not provide specific CAPEX or OPEX data .

Process Parameters and Conditions

The following table summarizes optimal process parameters reported across studies for effective bio-leaching.

Parameter	Optimal Range	Target Metal	Microorganism	Reference
Temperature	30℃	Cu	A. ferrooxidans	A. Santaolalla et al., 2021
Temperature	45℃	Au	Acidiplasma sp.	Intan Nurul Rizki et al., 2019
Temperature	60-80℃	Various	Thermophiles	Adegoke Isiaka Adetunji et al., 2024
pH	1.5-2.3	Various	Acidophiles	John Angel Sneha et al., 2023
pH	1.8	Cu	A. ferrooxidans	A. Santaolalla et al., 2021
pH	9	Au	P. balearica	Anil Kumar et al., 2018

Parameter	Optimal Range	Target Metal	Microorganism	Reference
Pulp density	1-10% w/v	Various	Various	Leidy Rendón-Castrillón et al., 2023
Pulp density	15 g/L	Cu, Zn	A. ferrooxidans-LR	Luciana Yamane et al., 2012
Duration	14-20 days	Cu, Au	A. niger / A. ferrooxidans	Multiple studies
Agitation	100-400 rpm	Various	Various	Leidy Rendón-Castrillón et al., 2023

Synthesis

Reconciling Heterogeneous Environmental Impact Findings

The substantial heterogeneity in reported environmental benefits of bio-leaching versus traditional methods can be explained through several mechanistic and contextual factors.

Population and Context Distinctions: Studies reporting dramatic carbon footprint reductions (60-99.99%) typically compare bio-leaching of secondary sources (WEEE) against primary ore extraction . For instance, the comparison showing 1.63 t CO2-e for recycling versus 10.8×10^6 t CO2-e for primary production represents a comparison between secondary source recycling and virgin ore mining—not a direct process-to-process comparison for the same waste stream. When comparing recycling technologies for the same waste stream, the environmental advantages of bio-leaching, while still significant (60% reduction in CO2), are more modest.

Study Quality Hierarchy: The most rigorous quantitative LCA data comes from a limited number of studies. Adegoke Isiaka Adetunji et al. (2024) provides specific CO2-equivalent values using established LCA methodology , while S. Balchandani et al. (2023) employed TRACI methodology and SimaPro software with defined system boundaries . Most other studies provide qualitative environmental benefit claims without supporting LCA data . This methodological disparity limits the ability to draw robust comparative conclusions.

Mechanistic Explanations for Performance Variability: The variability in metal recovery efficiency (e.g., gold recovery ranging from 42.5% to 98%) reflects fundamental differences in bio-leaching mechanisms:

- Autotrophic bio-leaching using iron- and sulfur-oxidizing bacteria operates optimally at low pH (1.5-2.3) and moderate temperatures (30-45℃), achieving high copper recovery but variable precious metal extraction
- Heterotrophic bio-leaching using organic acid-producing microorganisms (fungi, Gluconobacter) shows efficacy for REEs and precious metals at higher pH values
- Cyanogenic bio-leaching using Chromobacterium violaceum or Pseudomonas species enables gold dissolution at alkaline pH (9.0) but requires careful toxicity management

These mechanistically distinct approaches show complementary performance profiles rather than contradictory results.

Scale-Dependent Performance

A critical finding from this review is the inverse relationship between scale and data availability for environmental and economic metrics. Studies conducted at laboratory scale (100 mL to 10 L) provide the most detailed recovery efficiency data , while pilot-scale studies (50 kg to commercial) provide more relevant economic feasibility data but limited environmental impact quantification.

The technology readiness level assessment reveals that most bio-leaching processes remain at TRL 4 (laboratory validation) , with scalability challenges identified as a major barrier to commercial implementation . Specific challenges include:

- Maintaining efficiency at larger scales while managing heat dissipation and oxygen transfer
- Managing process stability with heterogeneous waste streams
- Achieving economic viability with extended processing times (14-203 days)

Economic Viability Considerations

The economic data suggests that bio-leaching can achieve competitive costs for gold recovery ($10,000–$20,000 USD/kg versus $30,000–$50,000 USD/kg for primary mining) , with positive NPV projections for industrial-scale operations (USD 127.4 million over 20 years with 65% IRR) . However, these projections are based on limited pilot-scale demonstrations and may not account for:

- High initial capital investment requirements (AUD 119.8-220 million)
- Extended processing times reducing throughput capacity
- Waste stream heterogeneity affecting process reliability

Studies consistently note that bio-leaching requires lower energy input and generates less toxic waste compared to pyrometallurgical approaches, but direct cost comparisons at equivalent scales remain scarce. The economic feasibility of bio-leaching appears most promising for:

- High-value metals (gold, PGMs) where recovery costs can be absorbed
- Waste streams with high metal concentrations (PCBs contain 40–800 times more gold than mined ore)
- Applications where environmental compliance costs favor low-emission technologies

Conclusions for Specific Applications

Based on the available evidence, the following conclusions can be drawn for specific contexts:

For copper recovery from PCBs: Bio-leaching using acidophilic iron-oxidizing bacteria (A. ferrooxidans) achieves recovery rates of 69-100% at laboratory scale , with demonstrated energy savings of approximately 90% . This represents the most mature application with the strongest evidence base.

For gold recovery from PCBs: Recovery efficiency varies substantially (42.5-98%) depending on the methodology . Thiourea-based bio-leaching shows promise (98% recovery) , while cyanogenic approaches achieve moderate efficiency (68-74%) . Combined approaches (bio-leaching followed by cyanidation) may optimize overall recovery .

For REE recovery: The evidence base is more limited, with recovery rates ranging from 9% for spent batteries to 65-100% for magnet swarf using engineered microorganisms . The use of lanthanophore-expressing bacteria shows promise for selectivity enhancement .

For PGM recovery: Limited studies report high recovery rates (90-94%) using biogenic cyanide approaches and living hydrogel systems , but these represent emerging technologies requiring further validation.

Critical data gaps: Comprehensive LCA studies with standardized methodology, system boundaries, and impact categories remain notably absent from the literature. Direct comparisons of CO2 emissions per kg of metal recovered, using consistent methodology across bio-leaching, pyrometallurgy, and hydrometallurgy for equivalent waste streams, have not been published. Economic feasibility assessments at pilot scale with detailed OPEX/CAPEX breakdown are available for only a handful of applications .

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What are the most efficient bio-leaching mechanisms for recovering Rare Earth Elements (REEs), Platinum Group Metals (PGMs), and Gold from Waste Electrical and Electronic Equipment (WEEE) using microbial consortia or biosynthesized nanoparticles? Specifically, I am looking for papers from 2019-2026 that discuss: 1. The use of 'Acidithiobacillus' and 'Aspergillus' strains or engineered microbial consortia. 2. The specific role of biosynthesized nanoparticles (e.g., bio-Pd, bio-Fe) in catalytic recovery. 3. Quantified recovery rates (percentage) and kinetics (time required). 4. Major bottlenecks identified, such as toxicity of WEEE components to the microbes or slow reaction times compared to chemical leaching.

The most efficient bio-leaching mechanisms for WEEE metal recovery combine acidolysis, redoxolysis, and complexolysis through microbial consortia, with thiourea bioleaching using Acidiplasma sp. achieving 98% gold recovery in 48 hours, Aspergillus niger consortia reaching 87% gold and 100% copper recovery, Acidithiobacillus-based systems consistently exceeding 90% copper extraction, biosorption with extremophilic bacteria attaining 87-99% REE adsorption, and Shewanella-based bio-Pd nanoparticle systems achieving approximately 90% palladium recovery—though widespread implementation remains constrained by WEEE toxicity to microorganisms, slower kinetics compared to chemical methods, and scale-up challenges.

Abstract

This systematic review of 40 sources (2011-2025) demonstrates that bio-leaching mechanisms for recovering valuable metals from WEEE vary substantially in efficiency depending on target metal, microbial system, and process conditions. For gold recovery, thiourea bioleaching with Acidiplasma sp. achieved the highest efficiency (98% within 48 hours) , while cyanogenic bacteria including engineered Chromobacterium violaceum and Bacillus megaterium strains showed moderate recovery (25-30%) . Aspergillus niger consortia demonstrated remarkable synergistic effects, with the MXPE6 + MX7 combination achieving 87% gold recovery compared to 17% for single strains . Copper recovery consistently exceeded 90% using Acidithiobacillus ferrooxidans and mixed acidophilic consortia within 7-14 days . REE recovery showed greater variability, with A. thiooxidans achieving 43% Pr and 28% Nd from LED lamps , while biosorption using Exiguobacterium sp. SH31 reached 87-99% adsorption capacity . For PGMs, living hydrogels containing Shewanella loihica PV-4 achieved approximately 90% Pd recovery with immediate process initiation , and biosynthesized Pd nanoparticles exhibited superior catalytic properties compared to commercial catalysts .

Major bottlenecks identified across studies include toxicity of WEEE components inhibiting microbial growth at elevated pulp densities , slow reaction kinetics (7-21 days) compared to chemical leaching , passive layer formation limiting REE extraction efficiency , and challenges in selective metal recovery from complex waste matrices . Scale-up remains a significant barrier, though successful demonstrations at 10 L scale with consistent yields have been reported . The evidence supports that acidolysis, redoxolysis, and complexolysis mechanisms operate synergistically in effective bio-leaching systems , with genetic engineering approaches enhancing lixiviant production and recovery efficiency . Bio-leaching offers documented environmental advantages over conventional methods , though economic viability depends on target metal value and process optimization for specific waste streams.

Paper search

We performed a semantic search using the query ”What are the most efficient bio-leaching mechanisms for recovering Rare Earth Elements (REEs), Platinum Group Metals (PGMs), and Gold from Waste Electrical and Electronic Equipment (WEEE) using microbial consortia or biosynthesized nanoparticles?

Specifically, I am looking for papers from 2019-2026 that discuss:

1. The use of 'Acidithiobacillus' and 'Aspergillus' strains or engineered microbial consortia.
2. The specific role of biosynthesized nanoparticles (e.g., bio-Pd, bio-Fe) in catalytic recovery.
3. Quantified recovery rates (percentage) and kinetics (time required).
4. Major bottlenecks identified, such as toxicity of WEEE components to the microbes or slow reaction times compared to chemical leaching.” across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 498 papers most relevant to the query.

Screening

Abstract screening

We screened in sources based on their abstracts that met these criteria:

- **Target Metals and Waste Source**: Does this study focus on bio-leaching or microbial recovery of rare earth elements (REEs), platinum group metals (PGMs), or gold from waste electrical and electronic equipment (WEEE)?
- **Biological Intervention Approach**: Does this study utilize Acidithiobacillus species, Aspergillus species, engineered microbial consortia, or biosynthesized nanoparticles (bio-Pd, bio-Fe, or similar) for metal recovery from electronic waste?
- **Quantitative Data Reporting**: Does this study report quantitative recovery rates (expressed as percentages) and/or kinetic data (time-based measurements)?
- **Study Type**: Is this study an experimental study, field study, pilot-scale study, systematic review, or meta-analysis?
- **Biological Component Inclusion**: Does this study include microbial or biosynthesized nanoparticle components (rather than focusing solely on non-biological chemical or physical metal recovery methods)?
- **Electronic Waste Component**: Does this study include electronic waste components (rather than focusing solely on metal recovery from other sources like ores, mining waste, or industrial sludge without WEEE)?
- **Target Metal Inclusion**: Does this study include at least one of the target metals (REEs, PGMs, or gold) rather than focusing exclusively on other metals?
- **Data Quality**: Does this study provide quantified recovery percentages or time-based measurements (rather than reporting only qualitative results)?
- **Publication Type**: Is this study a full research article, systematic review, or meta-analysis (rather than a conference abstract, editorial, opinion piece, or case report)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

427 papers passed abstract screening and moved to full-text screening.

Full-text screening

We then screened papers based on their full text using these additional criteria:

- **Target Metals and Waste Source**: Does this study focus on bio-leaching or microbial recovery of rare earth elements (REEs), platinum group metals (PGMs), or gold from waste electrical and electronic equipment (WEEE)?
- **Biological Intervention Approach**: Does this study utilize Acidithiobacillus species, Aspergillus species, engineered microbial consortia, or biosynthesized nanoparticles (bio-Pd, bio-Fe, or similar) for metal recovery from electronic waste?
- **Quantitative Data Reporting**: Does this study report quantitative recovery rates (expressed as percentages) and/or kinetic data (time-based measurements)?
- **Study Type**: Is this study an experimental study, field study, pilot-scale study, systematic review, or meta-analysis?
- **Biological Component Inclusion**: Does this study include microbial or biosynthesized nanoparticle components (rather than focusing solely on non-biological chemical or physical metal recovery methods)?
- **Electronic Waste Component**: Does this study include electronic waste components (rather than focusing solely on metal recovery from other sources like ores, mining waste, or industrial sludge without WEEE)?
- **Target Metal Inclusion**: Does this study include at least one of the target metals (REEs, PGMs, or gold) rather than focusing exclusively on other metals?
- **Data Quality**: Does this study provide quantified recovery percentages or time-based measurements (rather than reporting only qualitative results)?
- **Publication Type**: Is this study a full research article, systematic review, or meta-analysis (rather than a conference abstract, editorial, opinion piece, or case report)?

We considered all screening questions together and made a holistic judgement about whether to include each paper in the final analysis.

119 papers passed full-text screening and moved to data extraction.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Biological Agent**:

  Extract detailed information about the microbial system used including:

- Specific strain(s) or species name(s) (e.g., Acidithiobacillus ferrooxidans, Aspergillus niger MXPE6)
- Type of system (pure culture, mixed consortium, engineered consortia)
- Whether biosynthesized nanoparticles were used (bio-Pd, bio-Fe, etc.) and their role
- Any genetic modifications or engineering approaches
- Source/origin of the microorganisms (if mentioned)

- **Target Metals**:

  List all metals that were specifically targeted for recovery, focusing on:

- Rare Earth Elements (REEs) - specify which ones (La, Ce, Pr, Nd, Gd, etc.)
- Platinum Group Metals (PGMs) - specify which ones (Pt, Pd, Rh, etc.)
- Gold (Au)
- Other valuable metals recovered (Cu, Ni, Co, Ag, etc.) Note which metals were the primary focus versus secondary recoveries.

- **WEEE Source**:

Describe the electronic waste material used including:

- Type of WEEE (PCBs, motherboards, smartphones, hard drives, etc.)
- Specific components (gold fingers, integrated circuits, etc.)
- Preparation method (pulverized, shredded, particle size)
- Initial metal concentrations (if provided)
- Any pretreatment steps applied

- **Recovery Efficiency**:

  Extract quantitative recovery data including:

- Recovery percentages for each target metal (REEs, PGMs, Gold)
- Time required to achieve maximum recovery (kinetics)
- Leaching rates over time (if provided)
- Final metal concentrations in solution
- Comparison between different strains/conditions within the same study
- Statistical significance of results (if provided)

- **Processing Conditions**:

  Document the bio-leaching setup and conditions:

- Culture medium composition and pH
- Temperature and incubation time
- Solid-to-liquid ratio or pulp density
- Agitation/mixing conditions
- Scale of operation (lab, pilot, industrial scale)
- Sequential steps (if multi-stage process)
- Any chemical additives or organic acids used

- **Mechanisms Identified**:

  Extract information about how the bio-leaching works:

- Proposed biological mechanisms (acidolysis, redoxolysis, complexolysis)
- Role of metabolic products (organic acids, biosurfactants, etc.)
- Specific enzymes or biomolecules involved
- How nanoparticles enhance recovery (if applicable)
- Cellular uptake mechanisms
- Any mechanistic studies or characterization performed

- **Bottlenecks and Limitations**:

  Identify challenges and limitations reported:

- Toxicity effects of WEEE components on microorganisms
- Slow reaction rates compared to chemical leaching
- pH or temperature sensitivity issues
- Metal selectivity problems
- Scale-up challenges
- Economic viability concerns
- Environmental limitations

- Technical barriers to implementation
- **Performance Comparisons**:

  Extract any comparative data provided:

- Comparison to chemical leaching methods (efficiency, time, cost)
- Comparison between different microbial strains or consortia
- Comparison to other biotechnological approaches
- Advantages and disadvantages relative to conventional methods
- Economic analysis or cost-effectiveness data
- Environmental impact comparisons

Report

Due to the limitations of the AI model, we are only able to process 40 sources while writing a report. This report was written using the 40 sources that had the highest screening scores out of the 119 sources that we screened in and extracted data from.

Characteristics of Included Studies

The systematic review encompasses 40 sources published between 2011 and 2025, examining bio-leaching mechanisms for recovering valuable metals from electronic waste. The studies vary considerably in their focus, with some addressing primary experimental research on specific microbial systems while others provide comprehensive reviews of the field.

Study	Full text retrieved?	Study Type	Primary Target Metals	Microbial System	WEEE Source
Sahba Nadi et al., 2025	Yes	Primary study	REEs (Pr, Nd)	Acidithiobacillus thiooxidans	LED lamps
Svetlin Toshev et al., 2021	Yes	Primary study	Cu, Au	Mixed culture (A. ferrivorans, A. thiooxidans, others)	PCBs
C. García-Balboa et al., 2022	Yes	Primary study	Cu, Co, Al, Zn, REEs	Acidiphilium/Leptospirillum consortium + microalgae	Spent lamp powder
A. Brewer et al., 2019	Yes	Primary study	REEs (Nd, Pr, Dy)	Engineered E. coli with LBT	NdFeB magnets
Intan Nurul Rizki et al., 2019	Yes	Primary study	Au, Ag	Acidiplasma sp. Fv-Ap	PCBs
John Angel Sneha et al., 2023	Yes	Review	Au, Ag, Cu	A. thiooxidans, A. ferrooxidans, A. niger	PCBs, LCD screens
S. Tay et al., 2013	Yes	Primary study	Au	Engineered C. violaceum	PCBs

Study	Full text retrieved?	Study Type	Primary Target Metals	Microbial System	WEEE Source
Rosalba Argumedo-Delira et al., 2019	Yes	Primary study	Au	A. niger MXPE6 consortium	Mobile phone PCBs
Nathan M. Good et al., 2023	Yes	Primary study	REEs (Nd, Pr, Dy)	M. extorquens AM1	Pulverized smartphones
Mohammed Sami Al Sultan et al., 2023	Yes	Review	Au, Cu, REEs	A. ferrooxidans, A. niger, others	PCBs
David Kaplan et al., 2023	Yes	Primary study	REEs (Tb), PGMs (Pd)	Shewanella loihica PV-4	Not specified
E. Rosa et al., 2022	Yes	Primary study	Y, Cu, Pd	T. harzianum, P. glandicola, A. tubingensis	PC/mobile phone powder
Subhabrata Das et al., 2017	Yes	Primary study	Au, Cu	C. violaceum, D. acidovorans	Electronic scrap
Karem Gallardo et al., 2025	Yes	Primary study	REEs (Y, Pr, Nd, Gd, Tb, Dy)	Exiguobacterium sp. SH31	Spent mobile phones
A. Anz et al., 2018	Yes	Primary study	Au	Mutated B. megaterium	PCBs
Jorge Enrique Madrigal-Arias et al., 2015	Yes	Primary study	Au, Cu, Ni	A. niger MXPE6, A. niger MX7	Mobile phone PCBs, GFICMs
Nathan M. Good et al., 2023a	Yes	Primary study	REEs (Nd, Pr, Dy)	M. extorquens AM1 (engineered)	Nd magnet swarf, smartphones
A. Işıldar et al., 2018	Yes	Primary study	Cu, Au	Acidophilic and cyanogenic bacteria	PCBs from various devices
Éllen F. Rodrigues et al., 2024	Yes	Primary study	Cu, Au	A. niger	Computer PCBs
Chiara Magrini et al., 2025	Yes	Review	Cu, Au, Ag, REEs, PGMs	Multiple species including A. ferrooxidans, A. niger	Mobile phones
A. Jaiswal et al., 2025	Yes	Review	Cu, Au, Ag, Pt	A. ferrooxidans, A. thiooxidans, A. niger	Mobile phone PCBs
Anil Kumar et al., 2021	Yes	Primary study	Au, Cu	B. megaterium, L. sphaericus, C. violaceum	WPCBs

Study	Full text retrieved?	Study Type	Primary Target Metals	Microbial System	WEEE Source
Javad Aminian-Dehkordi et al., 2020	Yes	Primary study	Au, Cu	B. megaterium DSM319	Telecom PCBs
Nitish Kumar Bhol et al., 2025	Yes	Review	PGMs (Pt, Pd, Rh), Au, Ag	C. vulgaris, A. niger, P. chrysogenum	Not specified
Anna Hołda et al., 2021	Yes	Primary study	Cu, Al	A. ferrooxidans	Mobile phone WPCBs
F. Barmettler et al., 2016	Yes	Review	REEs (Y, La, Ce, Nd, others)	A. ferrooxidans, A. methanolicus, P. tricolor	Fluorescent powder
Juan Tapia et al., 2022	Yes	Primary study	Cu, Zn, Au	Tissierella, Acidiphilium, Leptospirillum consortium	PCBs
A. Murray et al., 2017	Yes	Primary study	PGMs (Pd, Pt)	E. coli MC4100	Spent automotive catalysts
P. Rasoulnia et al., 2021	Yes	Primary study	REEs, base metals	G. oxydans, S. pilosus	Spent NiMH batteries
Benjamin Monneron-Enaud et al., 2020	Yes	Primary study	Cu, Ag, REEs (Dy, Y)	A. ferrooxidans consortium	Medical PCBs
H. Garg et al., 2019	Yes	Primary study	Cu, Zn, Al, Ni	Mixed acidophilic consortium	Mobile phone PCBs
Sebastien Farnaud et al., 2019	Yes	Primary study	Cu, Zn, Mn	A. ferrooxidans	Telecom PCBs
S. De Corte et al., 2011	Yes	Review	PGMs (Pd)	Desulfovibrio, Shewanella, others	PCB leachates
Laura Castro et al., 2021	Yes	Review	REEs, Au, Cu	G. oxydans, A. niger, Y. lipolytica, A. ferrooxidans	PCBs, NdFeB magnets
M. Vítová et al., 2024	Yes	Review	REEs	A. ferrooxidans, A. thiooxidans, A. niger, others	Not specified
Pema Lhamo et al., 2022	Yes	Review	REEs	A. ferrooxidans, A. niger, P. tricolor	NdFeB magnets

Study	Full text retrieved?	Study Type	Primary Target Metals	Microbial System	WEEE Source
A. Sikander et al., 2022	Yes	Primary study	Cu, Ni, Fe, Zn, Al	Acidophilic consortium, A. niger	Mobile phone PCBs, tantalum capacitors
Helena Singer et al., 2023	Yes	Primary study	REEs (La, Ce, Pr, Nd)	M. fumariolicum SolV	Not specified
Y. Konishi et al., 2017	Yes	Primary study	PGMs (Pd, Pt, Rh), Au, In, Dy	Shewanella bacteria	Automotive catalysts, LCDs
A. Becci et al., 2020	Yes	Primary study	Cu, Zn	A. niger	Computer PCBs

The included studies demonstrate substantial diversity in methodological approaches. Primary experimental studies constitute the majority (n=28), while comprehensive reviews (n=12) provide valuable synthesis of the broader literature. The temporal distribution shows increasing research activity in recent years, with 32 studies published between 2019 and 2025, reflecting the growing interest in sustainable metal recovery technologies.

Recovery Efficiency of Target Metals

Gold Recovery

Gold recovery represents the most extensively studied application of bio-leaching from WEEE. Recovery efficiencies vary substantially depending on the microbial system employed, pretreatment conditions, and waste substrate characteristics.

Microbial System	WEEE Source	Recovery (%)	Time	Key Conditions
Engineered C. violaceum (pBAD)	PCBs	30%	8 days	0.5% w/v pulp density, 30℃
Wild-type C. violaceum	PCBs	11%	8 days	Standard LB medium
A. niger consortium	Mobile phone PCBs	56%	12-16 hours	No agitation, glucose medium
A. niger MXPE6 alone	Mobile phone PCBs	17%	32 days	No agitation
A. niger MXPE6 + MX7	PCBs	87%	14 days	280 rpm agitation
A. niger MXPE6 + MX7	GFICMs	28%	14 days	280 rpm agitation
Acidiplasma sp. Fv-Ap (TU-bioleaching)	PCBs	98%	48 hours	45℃, thiourea system
Mutated B. megaterium (pH 9.5)	PCBs	25%	Not specified	Alkaline conditions

Microbial System	WEEE Source	Recovery (%)	Time	Key Conditions
Wild B. megaterium (pH 7)	PCBs	11%	Not specified	Physiological pH
B. megaterium DSM319	Telecom PCBs	55%	4 days	Cysteine-supplemented medium
C. violaceum MTCC-2656	WPCBs	73.6%	7 days	pH 9, glycine supplementation
Acidophilic/cyanogenic bacteria	PCBs	44%	Not specified	Two-step bioleaching
A. niger (two-step method)	Computer PCBs	42.5%	14 days	2.5 g/L PCB waste
Iron-oxidizing consortium	PCBs	28%	11 days	Stirred tank reactor

The highest gold recovery (98%) was achieved using thiourea bioleaching with Acidiplasma sp. Fv-Ap , representing a departure from conventional cyanide-based approaches. This remarkable efficiency was attained within 48 hours , demonstrating that alternative lixiviant systems can substantially accelerate gold dissolution while reducing environmental concerns associated with cyanide.

Cyanogenic bacteria show considerable variation in gold recovery efficiency. Metabolically-engineered C. violaceum strains with duplicated hcnABC operons achieved 30% recovery compared to 11% for wild-type strains , demonstrating the potential of genetic engineering to enhance bioleaching performance. Similarly, mutated B. megaterium adapted to alkaline conditions (pH 9.5) showed enhanced gold recovery (25%) compared to unmutated strains at physiological pH (11%) .

Fungal consortia, particularly combinations of A. niger strains, demonstrate substantial potential for gold recovery. The combination of A. niger MXPE6 and A. niger MX7 achieved 87% gold recovery from PCBs , significantly outperforming single-strain approaches (17-42%) . This synergistic effect suggests that metabolic complementarity between strains enhances overall leaching efficiency.

Copper Recovery

Copper recovery from WEEE represents the most consistently successful application of bio-leaching, with multiple studies reporting efficiencies exceeding 90%.

Microbial System	WEEE Source	Recovery (%)	Time	Key Conditions
Mixed acidophilic consortium	Mobile phone PCBs	98-99%	Not specified	7-15% pulp density
A. ferrooxidans	WPCBs	90.23%	14 days	10% inoculum dose
Acidophilic/cyanogenic bacteria	PCBs	98.4%	Not specified	Two-step extraction
Acidophilic consortium	PCBs/tantalum capacitors	96%	10 days	Modified Silverman medium
A. niger	Computer PCBs	100%	14 days	Two-step method

Microbial System	WEEE Source	Recovery (%)	Time	Key Conditions
A. niger	PCBs	60%	21 days	Fe3+ as oxidant
B. megaterium DSM319	Telecom PCBs	85%	Several days	Cysteine supplementation
C. violaceum MTCC-2656	WPCBs	87.5%	7 days	pH 9 optimization
Acidophilic consortium	E-scraps	>99%	7 days	1% w/v pulp density
Iron-oxidizing consortium	PCBs	69%	11 days	Stirred tank reactor

Copper recovery consistently exceeds 90% across multiple microbial systems when optimal conditions are employed . The acidophilic consortium approach combining iron- and sulfur-oxidizing bacteria demonstrates particular effectiveness, achieving near-complete copper extraction (98-99%) even at elevated pulp densities of 10-15% . Complete copper recovery (100%) has been reported using A. niger in two-step bioleaching processes .

Rare Earth Element Recovery

REE recovery from WEEE represents an emerging application with considerable variability in reported efficiencies.

Microbial System	Target REEs	Recovery (%)	Time	WEEE Source
A. thiooxidans	Pr, Nd	43% (Pr), 28% (Nd)	3 days	LED lamps
M. extorquens AM1	Nd, Pr, Dy	65-100%	Not specified	Nd magnet swarf
Silk-based membranes (LBT)	Tb	Up to 85%	Not specified	Model solutions
Exiguobacterium sp. SH31	Y, Pr, Nd, Gd, Tb, Dy	87-99%	Not specified	Mobile phones
G. oxydans	Total REEs	9%	14 days	NiMH batteries
M. fumariolicum SolV	La, Ce, Pr, Nd	Up to 70%	Multiple cycles	Various sources
L. ferrooxidans	Er, Nd, Ce, La	100% (Er), 95% (Nd)	Not specified	Fly ash
A. niger	La	63%	Not specified	Various
Y. lipolytica	La, Ce, Nd	53% (La), 99% (Ce, Nd)	Not specified	Various
G. oxydans	REEs	Up to 56%	Not specified	Spent cracking catalysts

REE recovery shows substantial heterogeneity across studies, reflecting the diversity of microbial systems and waste substrates employed. Biosorption-based approaches using extremophilic bacteria show particularly promising results, with Exiguobacterium sp. SH31 achieving 87-99% adsorption capacity at pH 8 . The methylotrophic bacterium M. extorquens AM1 demonstrates 65-100% REE recovery in single process runs , with genetic engineering to eliminate exopolyphosphatase activity further enhancing metal accumulation .

The differential recovery of individual REEs is noteworthy. For instance, Pr recovery (43%) exceeded Nd recovery (28%) when using A. thiooxidans bioleaching of LED lamps , attributed to differences in ionic radii and the tetrad effect . Similarly, L. ferrooxidans showed complete extraction of Er (100%) but lower recovery of La (54%) from fly ash .

Platinum Group Metal Recovery

PGM recovery represents a specialized application of microbial systems, with particular emphasis on bio-nanoparticle formation.

Microbial System	Target PGMs	Recovery (%)	Application	Key Features
Shewanella loihica PV-4 (living hydrogel)	Pd	~90%	E-waste leachates	Immediate initiation
E. coli MC4100	Pd, Pt	Not quantified	Automotive catalysts	Bio-Pd/Pt nanoparticles
Shewanella bacteria	Pd, Pt, Rh	Not quantified	Automotive catalysts	Rapid reduction (<120 min)
Desulfovibrio/Shewanella	Pd	Not quantified	Various	Bio-Pd catalysis

The development of living hydrogels containing Shewanella loihica PV-4 achieved approximately 90% Pd recovery from simulated e-waste leaching streams . Notably, Pd recovery was initiated immediately after hydrogel introduction, compared to conventional biocarriers that require start-up times of hours to days . This represents a significant advancement in bioprocess efficiency.

Bio-palladium (bio-Pd) nanoparticle formation on bacterial cell surfaces provides dual functionality: metal recovery and catalyst production . Biogenic Pd nanoparticles exhibit superior catalytic properties compared to commercial Pd catalysts , enabling subsequent applications in environmental remediation and organic synthesis.

Bioleaching Mechanisms

Three primary mechanisms govern microbial metal solubilization from WEEE: acidolysis, redoxolysis, and complexolysis. Most effective bioleaching systems employ combinations of these mechanisms.

Acidolysis

Acidolysis involves metal dissolution through proton attack generated by microbial acid production. Autotrophic bacteria such as A. thiooxidans produce biogenic sulfuric acid through sulfur oxidation , while heterotrophic fungi including A. niger produce organic acids (citric, oxalic, gluconic) .

Acidithiobacillus thiooxidans achieved sulfuric acid concentrations of 10,500 mg/L (H+ 160 mM) through elemental sulfur oxidation , providing the acidic environment necessary for REE dissolution from LED lamps. The optimum inoculum volume of 5% (v/v) balanced nutrient availability against inhibitory effects at higher and lower concentrations .

Aspergillus niger produces multiple organic acids including oxalic, citric, and gluconic acids , with citric acid serving as the primary leaching agent for copper and zinc extraction . The production of organic acids was confirmed through FTIR spectroscopy . Gluconic acid production by G. oxydans reached 45.0 mM under optimized phosphate conditions , facilitating both acidolysis and complexolysis mechanisms.

Redoxolysis

Redoxolysis involves metal oxidation through microbially-mediated electron transfer reactions. Iron-oxidizing bacteria catalyze the oxidation of ferrous ions (Fe^{2+}) to ferric ions (Fe^{3+}), which subsequently oxidize metallic copper and other base metals .

The oxidation-reduction potential (ORP) serves as a critical process indicator in redoxolysis-based systems . Maintenance of elevated ORP through continuous bacterial iron oxidation ensures sustained metal leaching activity. In mixed acidophilic consortia, Leptospirillum species contribute chemolithotrophic iron oxidation capacity .

Shewanella bacteria employ reductive mechanisms for PGM recovery, reducing soluble Pd(II), Pt(IV), and Au(III) to insoluble metal nanoparticles at room temperature within 120 minutes . This bio-reduction process involves hydrogenase enzymes and cytochrome c3 , with hydrogen serving as the electron donor.

Complexolysis

Complexolysis involves metal solubilization through ligand complexation. Cyanogenic bacteria including C. violaceum and B. megaterium produce hydrogen cyanide as a secondary metabolite, which complexes gold into soluble $Au(CN)_2^-$ species .

Engineered enhancement of cyanogenesis has been achieved through metabolic engineering approaches. Integration of duplicated hcnABC operons under exogenous promoters increased cyanide production by 70% in C. violaceum , directly translating to improved gold recovery. Supplementation with cysteine (413 μM) in chemically-defined medium enhanced cyanide production by B. megaterium DSM319 .

Specialized ligands for REE complexation include lanthanophores and pyrroloquinoline quinone (PQQ) produced by M. extorquens AM1 . These REE-binding ligands enable selective bioleaching distinct from non-specific organic acid dissolution. Extracellular polymeric substances (EPS) produced by Exiguobacterium sp. SH31 contain hydroxyl and carbonyl functional groups that mediate REE binding through surface complexation .

Role of Biosynthesized Nanoparticles

Biosynthesized nanoparticles serve multiple functions in metal recovery systems. Bio-Pd nanoparticles formed on bacterial cell surfaces provide stable bionanocatalysts with enhanced catalytic activity compared to commercial alternatives .

The bio-Pd formation mechanism involves initial metal ion sorption followed by enzymatic reduction via hydrogenases . Nanoparticle deposition occurs primarily at the cell surface outer membrane, making particles accessible for catalytic applications . Pre-seeding cells with Pd(II) enhanced subsequent bimetallic catalyst formation, with 2-5 wt% Pd loading providing optimal catalytic activity .

Bio-reduced graphene oxide (B-rGO) serves as structural and functional linker in living hydrogel systems containing Shewanella loihica PV-4 , enhancing metal recovery while minimizing biomass production. This integrated approach demonstrated close to 90% Pd recovery with immediate process initiation .

Nanoparticle formation during fungal bioleaching has been observed with A. niger, producing metallic nanoparticles with spherical morphology and sizes varying according to PCB waste concentration . Gold nanoparticle precipitation by Delftia acidovorans provides an integrated recovery mechanism following cyanide leaching .

Bottlenecks and Limitations

Toxicity Effects on Microorganisms

WEEE components exert substantial toxicity on bioleaching microorganisms, representing a primary constraint on process efficiency. Inhibition of microbial activity by toxic compounds from PCBs has been documented across multiple microbial systems .

Toxicity manifests as reduced bacterial growth and diminished metabolite production. Higher pulp densities (>10%) lead to decreased metal bioleaching efficiency due to direct contact with toxic fractions . The toxic effect increases with PCB concentration, with lethal effects observed at higher pulp densities . Heavy metal ions inhibit A. niger growth and metabolism, while phenolic compounds from waste materials further suppress fungal activity .

Copper and other base metals compete with gold for cyanide ions during cyanogenic bioleaching , reducing gold recovery efficiency. Pretreatment with nitric acid to remove competing metals enhanced gold bioleaching by reducing cyanide consumption .

Higher e-scrap concentrations cause toxic effects on non-adapted bacterial cultures . Adaptation strategies including sequential exposure to increasing waste concentrations and pre-cultivation in stress conditions have been employed to enhance microbial tolerance.

Slow Reaction Kinetics

Bioleaching processes generally exhibit slower reaction rates compared to chemical leaching methods . Typical process times range from 7-21 days for complete metal extraction , compared to hours for chemical leaching .

Several factors contribute to kinetic limitations. Passive layer formation on particle surfaces hinders proton penetration during REE bioleaching , limiting extraction efficiency regardless of extended contact time. The controlled kinetics of cyanide-based gold leaching reflect both chemical reaction rate control and product diffusion limitations .

Process optimization through metabolic engineering and condition optimization has partially addressed kinetic constraints. Thiourea bioleaching with Acidiplasma sp. achieved 98% gold recovery within 48 hours , demonstrating that alternative lixiviant systems can substantially accelerate metal dissolution.

Scale-Up Challenges

Translation from laboratory to industrial scale remains a significant barrier for bio-leaching technology commercialization .

Scale-up has been demonstrated to 10 L bioreactors with consistent metal yields using M. extorquens AM1 , but larger-scale implementation faces additional challenges. Stirred tanks have high capital costs and energy demands, airlift percolators suffer from poor oxygen transfer and mixing, and column reactors face issues with uneven contact and prolonged reaction times .

Microbial biomass accumulation increases operational costs and process complexity at larger scales . Process integration strategies combining bioleaching with downstream separation and recovery steps require further development. The complexity and variability of WEEE composition necessitates flexible process designs adaptable to heterogeneous feedstocks .

pH and Temperature Sensitivity

Microbial systems exhibit narrow optimal ranges for pH and temperature, constraining operational flexibility .

Acidophilic bacteria require pH 1.5-2.5 for optimal activity , while low pH conditions can corrode equipment and negatively impact downstream processes . Extreme pH values (pH 3.6) during bioleaching present challenges for subsequent processing steps .

Cyanide availability for gold complexation is pH-dependent, with optimal extraction at pH 9-10 where HCN dissociates to CN- . However, most cyanogenic bacteria grow optimally at neutral pH , creating a process optimization challenge addressed through mutation or spent medium leaching at elevated pH .

Phosphate surplus in culture media can cause precipitation, particularly with REEs , limiting extractable metal fractions and complicating downstream processing.

Metal Selectivity

Selective recovery of specific metals from complex WEEE matrices remains challenging .

Non-specific leaching mechanisms dissolve both target and non-target metals, requiring additional separation steps . Organic acid production increases REE leaching but in a non-specific manner , necessitating engineered approaches for selective extraction. High concentrations of Cu and Fe can prevent REE binding to selective adsorbents .

Mixed metal compositions in WEEE present particular challenges. Difficulty in recovering certain metals like SnPb from complex alloys has been noted . Platinum seeding resulted in less effective bimetallic catalysts compared to Pd seeding , demonstrating metal-specific effects on bioprocess outcomes.

Economic Viability

Economic competitiveness remains uncertain for many bio-leaching applications .

Carbon substrate costs represent a major contributor to process economics . Low revenue from producing mixed rare earth oxides makes individual, separated REEs more desirable but more challenging to achieve . Current cyanide production levels may be insufficient for economically viable gold recovery .

Economic viability depends significantly on external factors including environmental regulations and resource recovery incentives . Full life cycle analysis is required to evaluate techno-economic feasibility .

Synthesis

Reconciling Variable Recovery Efficiencies

The substantial variation in reported recovery efficiencies reflects systematic differences in experimental conditions rather than inherent unreliability of bio-leaching approaches. Gold recovery ranging from 11% to 98% can be understood through several key variables.

Lixiviant system selection : Thiourea-based bioleaching achieved 98% gold recovery within 48 hours , substantially outperforming cyanide-based systems (11-30% in 8 days) . The thiourea system avoids competitive reactions with base metals while maintaining robust Fe-oxidation by Acidiplasma sp. . These findings suggest that lixiviant selection may be more important than microbial species for gold recovery.

Consortium synergy : Fungal consortia consistently outperform single strains. The A. niger MXPE6 + MX7 combination achieved 87% gold recovery compared to 17% for MXPE6 alone . This 5-fold enhancement reflects metabolic complementarity where different strains contribute distinct enzymatic activities and metabolites.

Genetic engineering impact : Engineered cyanogenic strains show 2-3 fold improvements over wild-type counterparts. C. violaceum with duplicated hcnABC operons achieved 30% versus 11% gold recovery , while alkaline-adapted B. megaterium mutants reached 25% versus 11% . These improvements directly correlate with enhanced cyanide production capacity.

Pretreatment effects : Removal of competing base metals through acid pretreatment enhanced gold recovery by reducing cyanide consumption . Two-step processes separating copper leaching from gold recovery optimize conditions for each metal.

Explaining Differential Metal Recovery

The observation that copper recovery (>90%) consistently exceeds gold recovery (~30-50%) across similar microbial systems reflects fundamental differences in metal chemistry and leaching mechanisms.

Copper dissolution proceeds through relatively straightforward acid attack and ferric oxidation , with the metal existing in accessible forms within PCB matrices. Multiple organic acids (citric, oxalic, gluconic) effectively solubilize copper . In contrast, gold requires specific complexing agents (cyanide, thiourea) for dissolution, with cyanide production being a secondary metabolite subject to regulatory constraints .

REE recovery efficiency varies by element due to ionic radii differences and the lanthanide tetrad effect . Pr recovery (43%) exceeded Nd recovery (28%) from the same LED lamp substrate , reflecting these physicochemical differences rather than process limitations.

Optimal Conditions by Application

The synthesis of experimental conditions across studies enables identification of application-specific optima:

For copper recovery : Acidophilic consortia at 1-10% pulp density, pH 1.8-2.0, 30℃, 7-14 days achieve >90% recovery . The iron-oxidizing pathway provides robust and reproducible performance.

For gold recovery : Two-step processes with base metal pretreatment, followed by cyanogenic leaching at pH 9-10, or thiourea bioleaching at 45℃ achieve optimal results . Fungal consortia in static culture offer a low-energy alternative with moderate efficiency .

For REE recovery : Biosorption approaches using extremophilic bacteria or engineered strains with lanthanide-binding proteins at pH 6-8 provide high selectivity . Methylotrophic bacteria utilizing waste-derived carbon substrates offer scalable alternatives .

For PGM recovery : Bio-reduction using Shewanella or related metal-reducing bacteria at neutral pH with hydrogen as electron donor enables simultaneous recovery and nanoparticle synthesis .

Addressing the Scale-Up Gap

The transition from laboratory to industrial scale requires addressing several interconnected challenges. Studies demonstrating 10 L scale-up with consistent yields provide proof-of-concept, but full commercialization requires:

1. **Process intensification** : Higher pulp densities (10-15%) have been successfully employed while maintaining >98% copper recovery, demonstrating that concentrated processing is feasible with appropriate microbial adaptation.
2. **Continuous operation** : Living hydrogel systems enabling immediate process initiation and column-based biosorption with reusable adsorbents offer pathways to continuous rather than batch operation.
3. **Integrated biorefining** : Sequential bioleaching followed by biouptake and biodismantling approaches demonstrate process integration strategies that maximize value recovery from complex waste streams.

Environmental and Economic Context

Bio-leaching offers documented environmental advantages including reduced energy consumption , elimination of harsh chemicals , and minimal hazardous emissions . The technology is consistently described as more environmentally friendly than pyrometallurgical and hydrometallurgical alternatives .

Economic viability depends on application-specific factors. High-value metals (gold, PGMs, separated REEs) offer favorable economics , while base metal recovery must compete with established low-cost chemical methods. The use of waste-derived carbon sources (milk whey , biodiesel glycerin) reduces substrate costs. Hybrid systems combining biological and chemical steps may optimize both efficiency and economics .

Regulatory trends favoring sustainable resource recovery and circular economy principles create increasingly favorable contexts for bio-leaching adoption. The technology readiness level varies by application, with copper bioleaching closest to commercial viability and REE recovery requiring further development.

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How does advanced pre-treatment and sorting of WEEE (Waste Electrical and Electronic Equipment) influence the efficiency of downstream bio-hydrometallurgical recovery of Critical Raw Materials? I need studies that focus on: 1. AI-driven spectral imaging (hyperspectral/multispectral) to identify specific CRM-rich components (like NdFeB magnets or high-grade PCBs). 2. The impact of removing toxic inhibitors (e.g., flame retardants, specific plastics) on microbial survival and leaching performance. 3. Automated robotic dismantling vs. shredding: benefits for bio-leaching yield.

Advanced pre-treatment and sorting of WEEE substantially improves bio-hydrometallurgical CRM recovery efficiency by enabling AI-driven spectral imaging to concentrate target elements up to 10,000-fold for subsequent high-purity extraction, removing toxic inhibitors that otherwise delay microbial activity by up to 2.6 times, and preserving CRM concentrations through selective automated dismantling that processes components 50 times faster than manual methods—with optimal pre-treatment selection dependent on initial CRM concentration, where high-value concentrated sources benefit most from robotic selective dismantling while bulk mixed waste streams achieve cost-effective base metal recovery through shredding combined with toxic fraction separation.

Abstract

This systematic review of 40 studies examining advanced pre-treatment and sorting of WEEE demonstrates that these upstream processes significantly influence downstream bio-hydrometallurgical recovery efficiency, though optimal approaches vary substantially by waste stream characteristics and target CRMs. AI-driven spectral imaging technologies, including machine vision and multi-energy X-ray transmission spectroscopy, achieved element enrichment up to 10,000 times original concentration , enabling neodymium recovery at 91.1% efficiency with 99.6% purity and optical sorting accuracies of 97.4% . Toxic inhibitor removal proved critical for microbial performance: plastics, ceramics, flame retardants, and heavy metals delayed microbial iron oxidation by up to 2.6 times , while pre-treatment strategies including electrostatic separation , ozonation , and preliminary acid leaching mitigated these effects. Automated robotic dismantling achieved 50-fold faster processing than manual methods (6 seconds vs. 5 minutes per HDD) , preserving CRM concentration in high-value components, whereas shredding to optimal particle sizes (0.5-1 mm) remained cost-effective for bulk base metal recovery, achieving 93-99% copper extraction .

The evidence indicates that pre-treatment selection should be guided by initial CRM concentration and downstream microbial processing parameters. For concentrated CRM sources like NdFeB magnets (27-32 wt.% REEs) , automated selective dismantling maximizes recovery value, while AI-driven sorting transforms previously uneconomic low-concentration CRM streams into viable recovery targets . For processes employing non-adapted microorganisms, toxic inhibitor removal is essential; however, serial subculturing can enable microbial adaptation to contaminated waste streams , potentially reducing pre-treatment requirements. The studies consistently demonstrate that component purity achieved through advanced sorting reduces chemical complexity in downstream processing , though head-to-head comparisons between pre-treatment approaches for identical waste streams and bio-hydrometallurgical processes remain limited in the literature.

Paper search

We performed a semantic search using the query "How does advanced pre-treatment and sorting of WEEE (Waste Electrical and Electronic Equipment) influence the efficiency of downstream bio-hydrometallurgical recovery of Critical Raw Materials?

I need studies that focus on:

1. AI-driven spectral imaging (hyperspectral/multispectral) to identify specific CRM-rich components (like NdFeB magnets or high-grade PCBs).
2. The impact of removing toxic inhibitors (e.g., flame retardants, specific plastics) on microbial survival and leaching performance.
3. Automated robotic dismantling vs. shredding: benefits for bio-leaching yield." across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 499 papers most relevant to the query.

Screening

Abstract screening

We screened in sources based on their abstracts that met these criteria:

- **WEEE Source Material**: Does this study focus on waste electrical and electronic equipment (WEEE) as the primary substrate or source material?
- **Critical Raw Materials Recovery**: Does this study target the recovery of materials classified as critical raw materials (e.g., rare earth elements, platinum group metals, lithium, cobalt, tantalum, indium, etc.)?
- **Advanced Pre-treatment Technologies**: Does this study employ at least one advanced pre-treatment method such as AI-driven spectral imaging (hyperspectral/multispectral), automated toxic inhibitor removal, or automated robotic dismantling systems?
- **Bio-hydrometallurgical Processing**: Does this study include biological leaching, bioleaching, or bio-hydrometallurgical processes as the recovery method following pre-treatment?
- **Quantitative Efficiency Measurements**: Does this study provide quantitative, measurable outcomes related to recovery efficiency, yield, selectivity, or process performance metrics?
- **Empirical Data**: Does this study present empirical data from laboratory experiments, pilot studies, field studies, or case studies (rather than being purely theoretical or computational modeling without experimental validation)?
- **Comparative Analysis**: Does this study include a comparative analysis component that compares different pre-treatment approaches or compares pre-treatment versus control conditions to assess impact on efficiency?
- **Methodological Detail**: Does this study provide sufficient methodological detail and information to allow for quality assessment (i.e., is it a full research article rather than just a conference abstract, poster, or brief communication without complete methodology)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

458 papers passed abstract screening and moved to full-text screening.

Full-text screening

We then screened papers based on their full text using these additional criteria:

- **WEEE Source Material**: Does this study focus on waste electrical and electronic equipment (WEEE) as the primary substrate or source material?
- **Critical Raw Materials Recovery**: Does this study target the recovery of materials classified as critical raw materials (e.g., rare earth elements, platinum group metals, lithium, cobalt, tantalum, indium, etc.)?
- **Advanced Pre-treatment Technologies**: Does this study employ at least one advanced pre-treatment method such as AI-driven spectral imaging (hyperspectral/multispectral), automated toxic inhibitor removal, or automated robotic dismantling systems?
- **Bio-hydrometallurgical Processing**: Does this study include biological leaching, bioleaching, or bio-hydrometallurgical processes as the recovery method following pre-treatment?
- **Quantitative Efficiency Measurements**: Does this study provide quantitative, measurable outcomes related to recovery efficiency, yield, selectivity, or process performance metrics?
- **Empirical Data**: Does this study present empirical data from laboratory experiments, pilot studies, field studies, or case studies (rather than being purely theoretical or computational modeling without experimental validation)?
- **Comparative Analysis**: Does this study include a comparative analysis component that compares different pre-treatment approaches or compares pre-treatment versus control conditions to assess impact on efficiency?
- **Methodological Detail**: Does this study provide sufficient methodological detail and information to allow for quality assessment (i.e., is it a full research article rather than just a conference abstract, poster, or brief communication without complete methodology)?

We considered all screening questions together and made a holistic judgement about whether to include each paper in the final analysis.

166 papers passed full-text screening and moved to data extraction.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Pre-treatment Technology**:

  Extract detailed information about the pre-treatment and sorting methods including:

- Specific technology type (AI-driven spectral imaging, robotic dismantling, manual sorting, shredding, etc.)
- Technical specifications (hyperspectral/multispectral parameters, automation level, sorting criteria)
- Process steps and sequence
- Equipment used and operational parameters
- Level of component separation achieved (whole components vs. crushed material)

- **WEEE Source**:

  Document the waste electrical and electronic equipment characteristics including:

- Type of WEEE (PCBs, hard drives, fluorescent lamps, etc.)
- Source and age of materials
- Initial composition and contamination levels
- Specific components targeted for separation (NdFeB magnets, high-grade PCBs, capacitors, etc.)
- Physical condition and preparation prior to treatment

- **Target CRMs**:

List all Critical Raw Materials specifically targeted for recovery including:

- Element names and chemical forms (REEs, platinum group metals, tantalum, etc.)
- Initial concentrations in waste stream
- Target components containing these CRMs
- Economic or strategic importance mentioned
- Purity requirements for recovered materials

- **Bio-hydrometallurgy Process**:

  Extract comprehensive details of the downstream recovery process including:

- Microorganisms used (species, strains, consortium composition)
- Leaching conditions (pH, temperature, oxygen levels, duration)
- Culture media and nutrients
- Reactor type and operational parameters
- Recovery method (precipitation, extraction, electrowinning, etc.)
- Process yields and selectivity

- **Toxic Inhibitor Impact**:

  Document effects of toxic compounds on bio-hydrometallurgical performance including:

- Specific inhibitors identified (flame retardants, plastics, heavy metals, etc.)
- Removal methods and effectiveness
- Impact on microbial survival, growth, and activity
- Changes in leaching performance with/without inhibitors
- Quantitative data on toxicity effects
- Mitigation strategies employed

- **Efficiency Metrics**:

  Extract all performance measures and outcomes including:

- Recovery rates/yields for target CRMs (percentages, concentrations)
- Processing time and throughput
- Purity of recovered materials
- Energy consumption and cost data
- Comparison metrics between different approaches
- Statistical significance of results
- Economic feasibility assessments

- **Method Comparisons**:

  Document any direct comparisons between pre-treatment approaches including:

- Specific methods compared (robotic vs. manual vs. shredding, etc.)
- Head-to-head performance data
- Advantages and disadvantages of each approach
- Conditions under which different methods perform better
- Combined or hybrid approaches and their performance
- Recommendations for optimal method selection

- **Mechanistic Insights**:

  Extract explanations for why certain pre-treatments improve bio-hydrometallurgical efficiency including:

- How pre-treatment affects surface area, accessibility, or liberation
- Impact on mass transfer and reaction kinetics
- Role of component purity vs. mixed material processing
- Contamination effects and their mechanisms
- Optimal particle size, composition, or preparation conditions
- Process bottlenecks and limiting factors identified

Report

Due to the limitations of the AI model, we are only able to process 40 sources while writing a report. This report was written using the 40 sources that had the highest screening scores out of the 166 sources that we screened in and extracted data from.

Characteristics of Included Studies

The review encompasses 40 studies examining the relationship between pre-treatment methods and bio-hydrometallurgical recovery of Critical Raw Materials from WEEE. The studies span multiple research designs, waste types, and target materials.

Study	Full text retrieved?	Study Type	Primary WEEE Source	Target CRMs	Pre-treatment Focus
M. Baniasadi et al. (n.d.)	Yes	Primary study	PCBs	REEs	Electrostatic separation
Benjamin Monneron-Enaud et al., 2020	Yes	Primary study	Medical PCBs	Dy, Y	Chemical pre-treatment, bio-dismantling
Luciana Yamane et al., 2012	Yes	Primary study	Computer PCBs	Cu, Zn, Al, Au	Magnetic/electrostatic separation
Seyi Rachel Dada et al., 2025	Yes	Primary study	Computer/mobile PCBs	Au, Ag, Cu, Zn, Nd	Manual dismantling, shredding
I. Birloaga et al., 2021	Yes	Primary study	PC/laptop WPCBs	Cu, Al, Au, Ag	Manual dismantling, semi-automated station
Z. Cherkezova-Zheleva et al., 2024	Yes	Review	NdFeB magnets	Nd, Pr, Dy, Tb	Microwave treatment, mechanochemistry

Study	Full text retrieved?	Study Type	Primary WEEE Source	Target CRMs	Pre-treatment Focus
Ange A. Maurice et al., 2021	Yes	Review	WPCBs	Ta, Nb, W, Ga, REEs	Multiple dismantling/sorting methods
J. J. Roy et al., 2021	Yes	Review	Spent LIBs	Li, Ni, Co, Mn	X-ray sorting, mechanical separation
Sandeep Panda et al., 2024	Yes	Primary study	LIBs and WPCBs	Co, Li, Cu, Ni	Shredding, NaOH treatment
Mohammed Sami Al Sultan et al., 2023	Yes	Review	PCBs, batteries, LCD panels	Li, Cu, Ni, REEs	Bio-dismantling
Juan Anaya-Garzon et al., 2021	Yes	Primary study	Low-grade PCBs	Au, Ag, Pd, Cu	Shredding
Jorge Enrique Madrigal-Arias et al., 2015	Yes	Primary study	Mobile PCBs, computer motherboards	Au, Cu, Ni	Manual dismantling
E. Rosa et al., 2022	Yes	Primary study	PC/mobile phone powder	Y, Cu, Pd	Mechanical grinding, microwave treatment
Diogo A. Ferreira-Filipe et al., 2025	Yes	Review	E-waste (general)	REEs	Not specified
Rosalba Argumedo-Delira et al., 2020	Yes	Review	Computer/cell phone PCBs	Au, Ag, Pd, Cu	Manual dismantling, magnetic separation
Jordi Vives Pons et al., 2024	Yes	Primary study	PCBs, copper cables	Cu, Fe	Shredding
A. Işıldar et al., 2018	Yes	Primary study	Discarded PCBs	Cu, Au, REEs	Not specified
Nathan M. Good et al., 2023	Yes	Primary study	NdFeB magnet swarf	Nd, Pr, Dy	Not specified
Svetlin Toshev et al., 2021	Yes	Primary study	PCBs	Cu, Au	Shredding and grinding
David Kaplan et al., 2023	Yes	Primary study	E-waste streams	REEs, PGMs	pH adjustment
C. Burkhardt et al., 2024	Yes	Primary study	Computer HDDs	NdFeB	Automated robotic dismantling
Anil Kumar et al., 2021	Yes	Primary study	WPCBs	Cu, Au	Shredding, pulverization

Study	Full text retrieved?	Study Type	Primary WEEE Source	Target CRMs	Pre-treatment Focus
V. Echavarri-Bravo et al., 2022	Yes	Primary study	EV LIBs	Mn, Co, Ni	Not specified
L. Yamane et al., 2020	Yes	Primary study	Computer PCBs	Au, Cu, Fe	Shredding, magnetic separation
V. Echavarri-Bravo et al., 2022a	Yes	Primary study	EV LIBs	Mn, Co, Ni	Not specified
Muhammad Mohsin et al., 2024	Yes	Primary study	WPCBs	Ta, Pd, Nb, Ga	AI-driven computer vision
Laura Castro et al., 2021	Yes	Review	Post-consumer devices	REEs	Not specified
E. A. Oke et al., 2024	Yes	Review	Discarded PCBs	Au, Ag, Pd, Ta	Smart disassembly, robotic dismantling
Dong Xia et al., 2024	Yes	Review	WPCBs	PGMs, REEs, Ta	AI-driven spectral imaging
Eva Benzal Montes et al., 2020	Yes	Primary study	Mobile phone PCBs	Cu	Manual dismantling, shredding
Nicolas M Charpentier et al., 2024	Yes	Primary study	PCBs	REEs, Ta, PGMs	AI-driven spectral imaging
Anil Kumar et al., 2018	Yes	Primary study	Waste PCBs	Au, Ag	Manual dismantling, pulverization
A. Becci et al., 2020	Yes	Primary study	Computer PCBs	Cu, Zn	Manual removal, shredding
A. Sikander et al., 2022	Yes	Primary study	Mobile phone PCBs	Ta	Manual dismantling, grinding
A. Sieber et al., 2025	Yes	Review	WEEE (general)	REEs, Ga	Not specified
M. Ueberschaar et al., 2017	Yes	Primary study	PCBs, HDDs, LCDs	Ga, In, REEs, Ta	Mechanical processing
Abhilash et al., 2021	Yes	Primary study	Shredded PCBs	Cu, Ni, Co	Shredding, physical separation

Study	Full text retrieved?	Study Type	Primary WEEE Source	Target CRMs	Pre-treatment Focus
Ewa Rudnik et al., 2025	Yes	Review	PCBs and LCDs	Sn	Mechanical dismantling, chemical methods
Nathan M. Good et al., 2023a	Yes	Primary study	NdFeB magnet swarf	Nd, Pr, Dy	Not specified
F. Coelho et al., 2021	Yes	Primary study	HDDs	Nd, Dy, Tb	Manual disassembly, thermal demagnetization

The studies demonstrate substantial heterogeneity in WEEE sources, with PCBs being the dominant waste stream investigated (28 studies), followed by lithium-ion batteries (5 studies), and hard disk drives containing NdFeB magnets (4 studies). Target CRMs span base metals (Cu, Zn), precious metals (Au, Ag, Pd), rare earth elements (Nd, Pr, Dy), and specialty metals (Ta, Ga, In). Pre-treatment approaches range from simple manual dismantling to advanced AI-driven sorting systems.

AI-Driven Spectral Imaging for CRM Identification

Technology Implementations

A subset of studies investigated advanced spectral imaging and AI-driven sorting technologies for identifying CRM-rich components prior to downstream processing.

Study	Technology Type	Technical Specifications	Sorting Accuracy/Performance	Target Components
Nicolas M Charpentier et al., 2024	Machine vision + MEXRT	128 energy levels (20-160 keV), 15 cm/s conveyor speed	Optical: 97.4%, MEXRT: up to 100%	CPUs, Ta capacitors, MLCCs
Muhammad Mohsin et al., 2024	YOLOv9/YOLOv11 deep learning	NVIDIA Jetson Nano, high-resolution camera	High mAP for component detection	ICs, capacitors, transistors
Dong Xia et al., 2024	Hyperspectral/multispectral with CNNs	X-ray transmission spectroscopy	Not quantified	Semiconductors, capacitors, magnetic devices
Ange A. Maurice et al., 2021	LIBS, XRF, machine vision	Laser pulses, broad X-ray beams	95% disassembly rate, >90% solder recovery	Ta, Nb, W, Ga, REEs

Study	Technology Type	Technical Specifications	Sorting Accuracy/Performance	Target Components
E. A. Oke et al., 2024	3D image recognition, machine vision	Robotic arms with suction grippers	75% separation success rate	Valuable ECs

The most comprehensive AI-driven sorting system reported by Charpentier et al. achieved element enrichment up to 10,000 times original concentration through combined optical and multi-energy X-ray transmission spectroscopy (MEXRT) sorting . This enrichment enabled subsequent recovery of neodymium at 91.1% efficiency with 99.6% purity . The optical sorting operated at 60 components per second while MEXRT processed 4 components per second .

Mohsin et al. demonstrated that AI-driven selective disassembly using YOLOv11 and RT-DETR models outperformed traditional methods in both efficiency and environmental impact . The system enabled real-time processing through integration with IoT and edge computing devices , with consistent improvement in model performance through iterative training .

The "dismantle and sort" strategy described by Dong Xia et al. was identified as more effective than "look and pick" approaches for large-scale recycling operations . This strategy employs AI-driven spectral imaging to maximize liberation and value extraction of electronic components .

Impact on Downstream Recovery Efficiency

The implementation of AI-driven sorting significantly impacts downstream bio-hydrometallurgical processes through component concentration. Charpentier et al. reported that their sorting approach simplified recycling by reducing stream complexity, which directly addresses bottlenecks related to heterogeneous electronic waste composition . The methodology enables tailored physico-hydrometallurgical processes with enhanced efficiency for recovering rare earth elements, tantalum, and platinum group metals .

Maurice et al. emphasized that sorting by composition increases elemental concentration, making recovery economically viable for materials present at low concentrations . The review noted that removing toxic inhibitors like flame retardants through sorting improves microbial survival and leaching performance .

Impact of Toxic Inhibitor Removal on Bio-Hydrometallurgical Performance

Identified Toxic Compounds and Their Effects

Multiple studies documented the inhibitory effects of various compounds present in WEEE on microbial activity during bioleaching.

Study	Toxic Inhibitors Identified	Impact on Microbial Activity	Quantitative Effects
M. Baniasadi et al. (n.d.)	Plastics, ceramics	Reduced accessibility, potential bacterial inhibition	Higher recovery in pre-treated samples
Juan Anaya-Garzon et al., 2021	Heavy metals (Pb, Ni), plastics	Delays microbial growth and iron oxidation	Pre-oxidation extended 2.6 times

Study	Toxic Inhibitors Identified	Impact on Microbial Activity	Quantitative Effects
Eva Benzal Montes et al., 2020	Ni, Cu, Al	Toxicity increases with concentration and time	Al inhibits at 0.5 M
Svetlin Toshev et al., 2021	Aromatics, phenols, polybrominated diphenyl ethers	Decreased bioleaching rate	Cu >90%, Au <50% recovery
J. J. Roy et al., 2021	Toxic metals, electrolytes (LiPF6), binders (PVDF)	Reduced microbial activity and survival	Decreased efficiency at high pulp density
A. Becci et al., 2020	Metal toxicity, PCB inhibition, phenols	Reduced citric acid production	Cu 60%, Zn 40% recovery
Ewa Rudnik et al., 2025	Tin ions	Delay in Fe2+ oxidation, inhibitory effect	Reduced leaching efficiency
David Kaplan et al., 2023	High Cu and Fe concentrations	Reduced REE recovery efficiency	~50% recovery with interfering metals present
V. Echavarri-Bravo et al., 2022	Ni at toxic concentrations	Affects bacterial activity and protein abundance	Reduction in protein abundance at 100 ppm Ni

The systematic study by Benzal Montes et al. demonstrated that metal concentrations previously reported as non-toxic became toxic when required leaching contact time was considered . This finding underscores the importance of kinetic considerations in toxicity assessment. Aluminum exhibited the most severe inhibitory effect at 0.5 M concentration .

Anaya-Garzon et al. quantified the impact of PCB toxicity on *Leptospirillum ferriphilum* , showing that elevated PCB loads extended the pre-oxidation phase by up to 2.6 times under batch conditions . However, successive subculturing reduced this delay, demonstrating adaptation potential .

Removal Methods and Mitigation Strategies

Study	Removal/Mitigation Strategy	Effectiveness	Performance Improvement
M. Baniasadi et al. (n.d.)	Electrostatic separation of non-metallic materials	Higher metal recovery	Improved recovery for most metals
Juan Anaya-Garzon et al., 2021	Serial subculturing for microbial adaptation	Reduced delay by up to 2.6 times	100% Cu, 100% Zn dissolution
Eva Benzal Montes et al., 2020	Separate biological and chemical oxidation steps	Mitigates toxicity	80% Cu recovery in 6 hours
Svetlin Toshev et al., 2021	Pre-establishing strong bacterial culture	Reduces inhibition	Cu >90%
A. Becci et al., 2020	Ozonation to reduce phenol concentration	Improved carbon source availability	Enhanced fungal metabolism
David Kaplan et al., 2023	pH adjustment and centrifugation	Partial effectiveness	Up to 85% REE recovery

Study	Removal/Mitigation Strategy	Effectiveness	Performance Improvement
Ewa Rudnik et al., 2025	Preliminary HNO3 leaching for Sn-Pb removal	Removes inhibitory solder	Improved downstream bioleaching
A. Sikander et al., 2022	Precultivation strategy	Mitigates inhibitory effects	99% Ni, 96% Cu extraction

The separation of metal fractions from plastics and flame retardants through mechanical pre-treatment improved material purity and microbial survival . Becci et al. further enhanced their process by using ozonation pretreatment to reduce phenol concentrations, improving carbon source availability for fungal metabolism in bioleaching with *Aspergillus niger* .

Benjamin Monneron-Enaud et al. demonstrated that pretreatment with 10 M NaOH removed protective coatings, increasing surface area and accessibility of copper circuits . The removal of toxic compounds like flame retardants was identified as beneficial for microbial survival and leaching performance .

Mechanistic Insights

The mechanistic basis for toxicity effects involves multiple pathways. Roy et al. identified that metal toxicity at high pulp densities reduces microbial activity and leaching efficiency through direct cellular damage . The optimal particle size in ultrafine and fine fractions improves metal recovery while minimizing toxic effects through better mass transfer .

Sieber et al. noted that pH is a critical parameter influencing biosorption by affecting functional group charges and metal speciation . Introduction of functional groups through chemical treatment can improve metal sorption capacity .

The paper by Ferreira-Filipe et al. highlighted that hazardous components including brominated compounds can impact microbial growth . The complexity of e-waste and presence of these hazardous components significantly affects bio-hydrometallurgical efficiency .

Automated Robotic Dismantling vs. Shredding

Comparative Performance Data

Study	Method Compared	Processing Time	Recovery/Performance	Key Advantages
C. Burkhardt et al., 2024	Automated robotic vs. semi-automated vs. manual	6 s vs. 2 min vs. 5 min per HDD	7.2 kg NdFeB/h capacity	50x faster than manual
Muhammad Mohsin et al., 2024	AI-driven selective vs. traditional	Real-time processing	High mAP detection	Superior efficiency, environmental impact

Study	Method Compared	Processing Time	Recovery/Performance	Key Advantages
Ange A. Maurice et al., 2021	Multiple methods compared	0.33 cm/s feeding rate	94% disassembling rate	Qualitative CAPEX/OPEX comparisons
E. A. Oke et al., 2024	Smart vs. mechanical vs. chemical	2 min 17 s per DPCB	75% separation success	Accurate component identification
Dong Xia et al., 2024	"Look and pick" vs. "dismantle and sort"	Not quantified	Not quantified	"Dismantle and sort" preferred for large-scale

The automated pilot plant described by Burkhardt et al. achieved magnet extraction in 6 seconds per HDD compared to 2 minutes for semi-automated and 5 minutes for manual dismantling . This represents a 50-fold improvement over manual methods. The system processes 600 HDDs per hour with annual capacity of 15 tonnes of NdFeB from 1.25 million HDDs . Power consumption remained below 7 kW .

Maurice et al. provided comprehensive comparisons of dismantling methods including manual dismantling, surface cutting, crude heating, infrared radiators, hot air heating, solder bath heating, and robotic dismantling . Each method exhibited distinct advantages: infrared radiators and hot air heating achieved high disassembly rates, while manual dismantling remained slow and labor-intensive . The review recommended combining methods like LIBS and computer vision for optimal sorting efficiencies .

Bio-Dismantling as an Alternative Approach

Benjamin Monneron-Enaud et al. introduced the concept of "biodismantling" using bioleaching technology to separate electronic components from PCBs . After 20 days of treatment with an iron-oxidizing mixed culture dominated by *Acidithiobacillus ferrooxidans* , successful dismantling was achieved . The process operated at pH 1.8 and 30℃ with 20 mM ferrous iron sulfate .

This biological approach was identified as a potential low-cost alternative to mechanical dismantling with lower energy consumption . Mohammed Sami Al Sultan et al. highlighted that bio-dismantling produces relatively high-grade components compared to grinding processes , reducing the need for subsequent separation steps.

Impact on Component Liberation and Bio-Leaching Yield

Study	Pre-treatment Method	Liberation/Accessibility Effect	Bio-leaching Yield
L. Yamane et al., 2020	Shredding (<2 mm) + magnetic separation	Concentrated metals, removed iron	99% Cu, 86% Au
Abhilash et al., 2021	Shredding (<150 μm)	Increased surface area	93% Cu (bacterial), 66% Cu (fungal)
Seyi Rachel Dada et al., 2025	Manual dismantling + shredding	Ferrous/non-ferrous separation	99% Cu, 95% Au
Eva Benzal Montes et al., 2020	Manual dismantling + shredding (<1 mm)	Controlled particle size 0.2-1 mm	80% Cu in 6 hours

Study	Pre-treatment Method	Liberation/Accessibility Effect	Bio-leaching Yield
A. Sikander et al., 2022	Manual dismantling + grinding (0.5-0.75 mm)	Optimal at 0.75 mm	99% Ni, 96% Cu

The relationship between particle size and bioleaching efficiency is well-documented. Sikander et al. determined that 0.75 mm represents an optimal particle size balancing surface area exposure against inhibitory effects . Smaller particles increase surface area but can form passivation layers or increase toxicity exposure.

Kumar et al. reported that lower pulp densities (10 g/L) with particle sizes ≤150 μm were more effective for metal mobilization due to reduced toxicity and better mass transfer . The study achieved 68.5% Au and 33.8% Ag recovery under optimized conditions .

Efficiency Metrics Across Studies

Metal Recovery Rates by Pre-treatment Approach

Study	Pre-treatment Category	Target Metal	Recovery Rate	Processing Time
Nicolas M Charpentier et al., 2024	AI-driven sorting	Nd	91.1%	Not specified
Seyi Rachel Dada et al., 2025	Hybrid mechanical	Au	95%	10 days + 6 hours
Seyi Rachel Dada et al., 2025	Hybrid mechanical	Cu	99%	10 days + 6 hours
L. Yamane et al., 2020	Shredding + magnetic	Cu	99%	15 days
L. Yamane et al., 2020	Shredding + magnetic	Au	86%	144 hours
Juan Anaya-Garzon et al., 2021	Shredding	Cu	100%	7 days batch, <48h continuous
Sandeep Panda et al., 2024	Shredding + NaOH	Co	98%	4 days optimum
Sandeep Panda et al., 2024	Shredding + NaOH	Li	80%	4 days optimum
Abhilash et al., 2021	Shredding	Cu	93%	10 days
A. Işıldar et al., 2018	Not specified	Cu	98.4%	50 min electrowinning
Nathan M. Good et al., 2023a	Not specified	Nd	65-100%	Scalable to 10 L

Purity and Economic Considerations

Study	Metal/Material	Purity Achieved	Economic Assessment
Nicolas M Charpentier et al., 2024	Nd oxide	99.6%	Capital recoverable in 3-5 years
Seyi Rachel Dada et al., 2025	Cu	>99%	Reduced chemical costs
Seyi Rachel Dada et al., 2025	Au	>70%	Economic viability demonstrated
Abhilash et al., 2021	Cu sulfate	>99%	More cost-effective than conventional
E. A. Oke et al., 2024	Cu deposits	>99.9%	Low energy consumption
A. Işıldar et al., 2018	Cu	65%	Techno-economic assessment conducted

The AI-driven sorting approach described by Charpentier et al. demonstrated superior sustainability with capital expenditures recoverable within three to five years . The process achieved 99.6% purity for neodymium oxide , enabling direct use in magnet manufacturing without additional refining.

Jordi Vives Pons et al. reported that bio-based technology significantly reduces energy consumption and emissions compared to traditional methods . Their gradient boosting model achieved 84% accuracy in predicting copper recovery from RGB sensor data .

Synthesis

Reconciling Heterogeneous Findings

The studies examined reveal substantial variation in pre-treatment approaches and their effects on bio-hydrometallurgical recovery efficiency. This heterogeneity can be explained through several key factors:

Waste Stream Complexity and Target CRMs

Studies targeting rare earth elements from NdFeB magnets (Burkhardt et al., Good et al., Charpentier et al.) consistently demonstrate that selective component separation preserves CRM concentration, enabling higher recovery efficiencies . In contrast, studies processing mixed PCB waste through shredding (Anaya-Garzon et al., Yamane et al.) achieve high recovery rates for base metals but face challenges with precious metals and REEs present at lower concentrations .

The distinction lies in initial CRM concentration: NdFeB magnets contain 27-32 wt.% REEs , while PCBs typically contain <0.1% critical metals . This concentration difference fundamentally influences the value proposition of advanced sorting versus bulk processing.

Microbial Tolerance and Adaptation

Studies reporting high recovery rates with minimal pre-treatment (Anaya-Garzon et al., Sikander et al.) employed microbial adaptation strategies to overcome toxicity . Successive subculturing boosted microbial activity by reducing inhibition delays by up to 2.6 times . In contrast, studies without adaptation protocols reported significant toxicity-related performance limitations .

This suggests that the value of toxic inhibitor removal through pre-treatment depends substantially on downstream microbial processing parameters. Processes using adapted microorganisms may tolerate higher contaminant levels, reducing the necessity for extensive pre-treatment.

Scale and Economic Context

The automated robotic dismantling system described by Burkhardt et al. achieved 6-second processing per HDD but requires substantial capital investment . This approach is economically justified for concentrated REE sources (NdFeB magnets) but may not be viable for lower-value waste streams. Conversely, shredding-based approaches remain cost-effective for bulk PCB processing despite lower selectivity .

Charpentier et al. demonstrated that AI-driven sorting can achieve economic viability with 3-5 year capital recovery through element enrichment up to 10,000 times original concentration . This enrichment factor transforms marginal CRM sources into economically recoverable streams.

Optimal Pre-treatment Selection Framework

Based on the evidence synthesis, optimal pre-treatment selection depends on:

1. **For high-value, concentrated CRM sources (NdFeB magnets, high-grade components)**: Automated robotic dismantling preserves component integrity and CRM concentration, enabling superior downstream recovery . Processing times of 6 seconds per unit with >90% recovery efficiency justify higher capital investment .
2. **For mixed WEEE streams with diverse CRMs**: AI-driven spectral imaging and sorting achieves element enrichment enabling economic recovery of previously unrecoverable materials . Combined optical and MEXRT sorting provides optimal balance of speed and accuracy .
3. **For bulk base metal recovery**: Shredding followed by mechanical separation remains cost-effective, achieving 93-99% copper recovery . Toxic inhibitor removal through separation of plastic and flame retardant fractions improves microbial performance .
4. **For processes using non-adapted microorganisms**: Pre-treatment to remove toxic inhibitors is essential. Separation of non-metallic materials improved recovery rates , while ozonation reduced phenol toxicity .

Mechanistic Basis for Pre-treatment Effects

The mechanistic pathways through which pre-treatment influences bio-hydrometallurgical efficiency include:

Surface Area and Accessibility: Particle size reduction through shredding increases surface area available for microbial attack . However, excessively fine particles (<0.5 mm) can increase toxicity exposure and form passivation layers . Optimal particle size ranges from 0.5-1 mm for most applications .

Component Purity: AI-driven sorting increases component purity, reducing chemical complexity and enabling tailored recovery processes . This simplification addresses bottlenecks associated with heterogeneous waste composition .

Mass Transfer Enhancement: Pre-treatment affects mass transfer kinetics through particle size optimization and removal of barriers to reagent access . The bio-generated acidic environment in bioleaching mirrors natural ore processing but requires accessible metal surfaces .

Toxicity Mitigation: Removal of flame retardants, heavy metals, and organic compounds protects microbial viability . Serial adaptation provides an alternative strategy when complete inhibitor removal is impractical .

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